CHAPTER 11

Restoration of Broken Symmetries

11.1 Introduction

In the investigation of the properties of the nuclear interaction (Chap. 4), we showed that the exact many-body Hamiltonian H is invariant under a number of symmetry operations, that is, it commutes with the corresponding symmetry operator S:

$$[H, S] = 0. \tag{11.1}$$

Therefore, we can always find wave functions that are simultaneous eigenfunctions of H and S. Examples are the three components of the linear momentum $\mathbf{P}$, the particle number N, and the angular momentum operators $\mathbf{J}^2$, J_z.

We have also seen that this general property is very useful in cases of a few particles, where it can be used to reduce the size of the corresponding eigenvalue problem dramatically by working in appropriate eigenspaces of the symmetry operators. The elimination of the center of mass coordinate in a two-body system or angular momentum coupling techniques applied for a few particles outside a closed shell are examples.

In the general case of many particles and strong correlations, the proper treatment of symmetries is a serious problem: On the one hand, we want to describe the system by *simple wave functions*, such as product states of independently moving particles (or quasi-particles); on the other hand, we are not able to take into account *important correlations* between the nucleons by such simple wave functions if we require simultaneously the

proper symmetry behavior. The only translational invariant product wave functions, for example, are products of plane waves. They are certainly not able to describe the strong correlations among the nucleons, which cause their clustering into a finite nucleus.

In Chapter 2 we saw, however, that such correlations can be described by a mean field (the shell model) and that product states of shell model wave functions provide a very reasonable description of many nuclear properties. Such wave functions, however, break the translational symmetry.

It is a general procedure,* which is very important in nuclear physics, that correlations are treated by a *symmetry-violating mean field approach.* Whether such a description is possible and is a good approximation certainly depends on the nature of the correlations. The long-range particle–hole correlations which cause stable deformations, and the *pp*-correlations, which provide superfluid properties, are in any case of the mean field type. In Chapter 7 we saw that they can be treated by the HFB theory, a mean field approach that violates angular momentum and particle number. It turns out that the stronger the correlations, the better such an approximation. In analogy to solid state physics, we say that the system undergoes a *phase transition* to a symmetry-violating state, as, for instance, to a deformed state or to a superfluid phase. Of course, classical and macroscopic considerations enter our arguments when we say the nucleus is localized, is deformed, or has a superfluid phase. We will see later in this chapter that the mean field description of phase transition eventually becomes exact in very large (macroscopic) systems. This means that for certain measurable quantities it may be irrelevant as to whether we calculate them in the symmetry-broken mean field approach or in a symmetry-nonbreaking theory, which of course is, in principle, required by quantum mechanics. Imagine a droplet of the quantum fluid ^{3}He with various numbers of particles. Once the drop is so big that it can actually be "seen," it is certainly an exact description to consider the drop (or its mean field) as localized (wave paket), though quantum mechanics would still require a translationally invariant treatment for stationary states.

Nuclei are, of course, not macroscopic objects, but the heavier ones are big enough that a mean field treatment—which for a macroscopic drop would eventually become exact—is already a very good description for certain quantities. These quantities are, of course, those which are also macroscopically defined as, for instance, radius, deformation,† superfluid current, ground state energy, etc. Those quantities which are, however, of a quantum nature, as, for example, discrete energy spectra and transitions

* For a more detailed discussion of this point, see [Li 60a].

† The deformation of a nucleus is, of course, not a directly measurable quantity as in a macroscopic system, but it can be defined in a model-dependent way only rather indirectly over the "inner" quadrupole moment (see Secs. 1.5.2 and 11.4.6.4); once this relationship is established once and for all, the mean field theory also reproduces the nuclear deformation rather well over the whole periodic system.

between excited states, will not be describable in a pure mean field approach.

In particular, the product wave function is a very poor approximation of the exact many-body state. As we shall see, it has been tried to interpret the symmetry-violating wave function as a function in an "*intrinsic*" *coordinate frame* moving with the nucleus. All quantities which depend only on this internal function are eventually reproduced quite well in the mean field approach.

There are, however, two reasons to go beyond the mean field approach:

(i) In nuclear physics we are also interested in quantities such as transition probabilities and electromagnetic moments, which can never be calculated from a symmetry-violating Nilsson wave function alone.

(ii) The nucleus is a finite system. The phase-transition is therefore always smeared out. We often find a gradual transition from conservation to weak violation, and eventually a strong breaking of the symmetry. In cases of weak symmetry violation we have to go beyond the mean field approach and incorporate the symmetries properly.

In Section 11.2 we wish first to discuss in more detail the mechanism of symmetry breaking in the mean field approach. It gives us a better understanding of the methods used in Chapters 5, 6 and 7. We present this discussion here because in those chapters we did not yet have the techniques of Chapters 9 and 10.

To *restore the symmetries* broken in the mean field approach, there have been attempts to transform explicitly to an "*intrinsic*" *system*. Such attempts, however, are often intimately connected with basic difficulties. On the other hand, such ideas are used in many phenomenological models (see Sec. 1.5). We therefore discuss them briefly in Section 11.3. The usual means to develop a symmetry-conserving theory on the basis of wave functions obtained from a mean field approach is provided by *projection techniques* onto eigenspaces of the symmetry operator. These are discussed in Section 11.4.

We should also mention the RPA approximation, which goes beyond the static mean field approach. It includes higher correlations and provides an approximate restoration of the symmetry (see Sec. 8.4.7 and [UW 65, GW 68, MW 69b, MW 70, Bi 76, Ma 77a]). As we shall see in Section 12.3.2, we can also derive this method in a time-dependent mean field theory, which shows that this more general mean field approach can already restore some aspects of symmetry violation.

In this chapter we treat mainly the symmetry violations of *translational invariance* (transition to a *localized nucleus*), of *particle number* (phase transition to a *superfluid state*), and *angular momentum* (phase transition to a *deformed state*). There are, of course, also other symmetries, like the parity [BSB 69] (which is also broken in the context of pion condensation

[MKS 74, BF 74, BCD 75, BW 76]) or the isospin (which is only an approximate symmetry [GW 68]). They can be handled in a very similar fashion.

Investigations have also been carried out on the possibility of a crystalline structure of nuclear matter with a periodic long-range order described by non-plane-wave HF-states, which violate translational symmetry (Overhauser states [Ov 60, ALP 77, AAZ 77]). One finds that for realistic nuclear forces such wave functions generally yield a higher energy than plane waves. Only at very low subnuclear densities can a stable crystalline structure of α particles develop (see also [BC 73]).

Symmetry violations discussed in this chapter are based on the mean field approach. Certainly, there are also other symmetry breaking theories. They can sometimes be treated with similar techniques.

11.2 Symmetry Violation in the Mean Field Theory

As we saw in Section 4.3 and Chapter 7, in the nucleus there are very different correlations: On the one hand, there are the very short-range Brueckner correlations, which have nothing to do with symmetry violations. On the other hand, there are long-ranged correlations caused by *ph* and *pp* forces.* If they are weak, they can be treated by the RPA approximation and yield harmonic collective vibrations and a correlated ground state. If they become stronger we find anharmonicities and modes that are lowered drastically in energy (soft modes). The corresponding energy surface becomes very flat in one direction and, with sufficiently strong correlations, a sudden symmetry-violating minimum develops. In such cases the mean field approximation discussed in Chapter 7 yields non-vanishing deformation potentials Γ or pairing fields Δ as solutions of the nonlinear "gap equations" (7.41)

$$\Gamma_{lm} = \sum_{rs} v^{ph}_{lsmr}\rho_{rs}, \qquad \Delta_{lm} = \sum_{rs} v^{pp}_{lmrs}\kappa_{rs} \tag{11.2}$$

where v^{ph} and v^{pp} are the effective forces in the *ph* and the *pp* channel and the densities ρ and κ [Eq. (7.22)] depend nonlinearly on Γ and Δ through the diagonalization of the HFB equations (7.42).

Since the densities ρ and κ are calculated with symmetry-breaking wave functions, they have a monopole part ρ_0, κ_0, a quadrupole part ρ_2, κ_2, and so on. They are obtained from coupling the indices r and s to angular momentum $I = 0, 2, \ldots$. ρ_0 corresponds to a spherical density distribution and yields the spherical part V_0 of the potential Γ. It is well pronounced for all nuclei and corresponds to the spherical shell model potential. In the following we shall work in this basis.

* We have to emphasize, however, that this does not mean that the range of the forces that produce these correlations is large.

Depending on the strength of the interactions v^{ph} and v^{pp} and on the level density in the vicinity of the fermi surface, the "gap" equations (11.2) may have non-vanishing solutions also for the quadrupole part Γ_2, the monopole pairing field Δ_0, and eventually for higher terms, which are less important in nuclei.* We can therefore divide the mean field into three parts, V_0, Γ_2, and Δ_0. They are connected with the violation of translational symmetry, rotational symmetry, and particle number.

In Section 4.4.7 we saw that we can decompose the interactions v^{ph} and v^{pp} into components with different angular momentum. According to Eqs. (4.124) and (4.131), they are separable in the angles. Since the radial dependence only enters at the vicinity of the nuclear surface, we can replace it to a good approximation by a constant and get for the quadrupole part of v^{ph}, and from the monopole part of v^{pp} the well known pairing-plus-quadruple model (see Sec. 7.4), which we shall use for the following discussion of phase transitions in the nucleus.

We first investigate the *quadrupole correlations*. In this case, the *effective* two-body Hamiltonian leading to the one body equation (11.2) can be written in the form

$$H=\sum_k \epsilon_k c_k^+ c_k - \frac{\chi}{2}\sum_\mu Q_\mu^+ Q_\mu; \qquad \chi>0. \tag{11.3}$$

c_k^+, c_k are particle operators in the spherical shell model basis and ϵ_k are the corresponding eigenvalues of $T+V_0$. The quadrupole operators Q_μ are given by

$$Q_\mu=\sum_{k_1 k_2}\langle k_1|r^2 Y_{2\mu}|k_2\rangle c_{k_1}^+ c_{k_2}. \tag{11.4}$$

The "deformation" equation (11.2) now has the form (7.75) and determines the deformation parameters q_μ:

$$q_\mu=\chi\langle\Phi|Q_\mu|\Phi\rangle. \tag{11.5}$$

To avoid the problem of HF in open shells we furthermore assume that for vanishing interaction ($\chi=0$) the spherical subshells i with $\epsilon_i \leqslant \epsilon_F$ are occupied and the subshells ϵ_m with $\epsilon_m > \epsilon_F$ are empty.

For non-vanishing interactions the behavior of the system depends very much on the relation of the average level spacing in the vicinity of the fermi surface† to strength χ. For small level density, for instance, for magic nuclei, or for small values of χ we have the case of spherical nuclei. The "deformation" equation (11.5) only has the solution $q_\mu=0$. It corresponds to a minimum in the energy surface and we can calculate vibrational states with the frequency Ω in the framework of the RPA [Eq. (8.135)]:

$$\frac{1}{\chi}=\sum_{mi}\left|\langle m\|r^2 Y_2\|i\rangle\right|^2\frac{2(\epsilon_m-\epsilon_i)}{(\epsilon_m-\epsilon_i)^2-(\hbar\Omega)^2}. \tag{11.6}$$

*For instance, octupole deformations (Γ_3) or quadrupole pairing (Δ_2); see also the multipole expansion in Sec. 4.4.7.

†See the analogous discussion of Δ in Section 6.3.5.

With increasing level density (i.e., if we go to nuclei far from closed major shells) or for increasing strength χ we come to a point where the RPA energy goes to zero, which is equivalent to the fact that the spherical solution $q_\mu = 0$ of Eq. (11.5) no longer corresponds to a minimum in the energy surface (see Sec. 8.4.3). There exists at least one deformed solution $|\Phi_0\rangle$ of Eq. (11.5), with

$$\langle\Phi_0|Q_\mu|\Phi_0\rangle \neq 0 \qquad \text{for at least one } \mu. \tag{11.7}$$

It violates rotational symmetry. We are now able to define an *intrinsic coordinate system*, for instance, by the principal axis of the mass distribution in the function $|\Phi_0\rangle$. At the same time we see that there is now a whole set of solutions $|\Phi(\Omega)\rangle$ to the HF equations that are obtained from $|\Phi_0\rangle$ by an arbitrary rotation in space characterized by the Euler angles $\Omega = (\alpha, \beta, \gamma)$ [see Eq. (A.4)]

$$|\Phi(\Omega)\rangle = \hat{R}(\Omega)|\Phi_0\rangle \tag{11.8}$$

as it is schematically shown in Fig. 11.1.

Because of the rotational invariance of the Hamiltonian, all these solutions have the same energy

$$\langle\Phi(\Omega)|H|\Phi(\Omega)\rangle = \langle\Phi_0|H|\Phi_0\rangle. \tag{11.9}$$

In particular, this means that the deformed state does not correspond to a minimum but rather to a point of the bottom of a valley in the energy surface. In Section 11.4 we will see that this degeneracy in energy can be used to construct a linear superposition of all these wave functions $|\Phi(\Omega)\rangle$, which is then an eigenfunction of the symmetry operators $\mathbf{J}^2$ and J_z.

The transition from spherical nuclei in the vicinity of closed major shells to deformed nuclei in the middle of major shells is often called a *phase transition*. This is analogous to a macroscopic system, where with increasing correlations we have the same mechanism: a soft mode connected to the violation of a symmetry. There are, however, two important differences between a usual phase transition in solid state theory and a "phase transition" in nuclear physics:

(i) In a macroscopic system the transition is strictly *discontinuous*, that is, the different quantities (order parameters) change abruptly at a certain value of a continuous parameter. For a finite system the

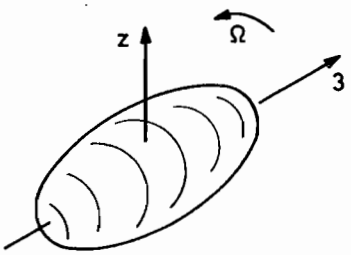


Figure 11.1. Schematic representation of the fact that the mean field approach defines an internal coordinate frame.

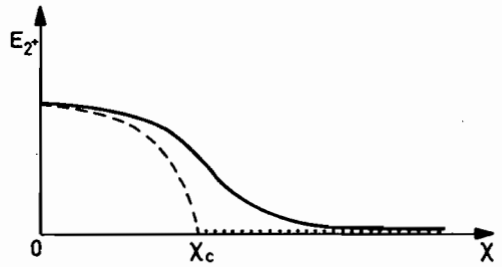


Figure 11.2. Schematic representation of the lowest excitation energy as a function of quadrupole coupling strength: exact solution (full line) and RPA approach in the spherical basis (dashed line) and in the deformed basis (dotted line). In the latter case, the lowest RPA solution is the spurious state, which corresponds to the rotation (see Sec. 8.4.7).

sharp transition is washed out and a more or less *gradual change* in the order parameters given by the deformations q_μ (or the gap Δ) is observed. This fact is somewhat hidden if our procedure is followed: If we calculate vibrations around a spherical shape in RPA we find that with increasing coupling strength χ, one frequency goes to zero at a sharp critical value χ_c. For all values $\chi > \chi_c$ we find a deformed solution. However, in a finite system both methods—the RPA in the spherical case and the deformed HF-method in the deformed case—break down in the vicinity of the critical point and do not provide a method for describing the transition quantitatively. In an exact solution, as shown schematically in Fig. 11.2, we find that the vibrational mode is lowered dramatically in energy in the transition region and goes over into the 2^+ rotational state in the deformed region, which goes to zero only in the limit of very large deformations or strong correlations.

(ii) Whereas in the macroscopic case the mean field approximation can provide an *exact description* of many quantities, this is no longer the case in the finite nuclear system. As we shall see in Section 11.4.4, the approximation becomes good only in the limit of strong symmetry violations. There are many nuclei in the transition region, where a static mean field approach is not applicable. From that point of view, the three symmetry violations of linear momentum, angular momentum, and particle number are very different. The quality of the mean field approximation is very good in the translational case for all medium and heavy nuclei. In the rotational case it is good only for the well-deformed regions. However, we also observe spherical nuclei without rotational symmetry violations; there are also large transitional regions, where the mean field approach breaks down. BCS correlations in nuclei are usually quite weak, therefore the corresponding mean field approximation, the BCS theory, is often not good enough to reproduce the experimental data. We shall therefore investigate, in the following sections, methods of improving the deficiencies of the mean field approximation.

An *experimental measure* of the strength of the quadrupole correlations, and therefore also an indication of the validity of the mean field approximation, is given by the *fluctuation of the quadrupole operator* in the exact wave function $|\Psi\rangle$

$$\Delta Q^2 = \langle\Psi|Q\cdot Q|\Psi\rangle - \langle\Psi|Q|\Psi\rangle\cdot\langle\Psi|Q|\Psi\rangle. \tag{11.10}$$

(The dot means a scalar product of the five quadrupole operators.) Since the expectation values $\langle\Psi|Q_\mu|\Psi\rangle$ vanish in the ground state with $I=0$, we find that ΔQ^2 is proportional to the quadrupole correlation energy of the Hamiltonian (11.3)

$$\Delta Q^2 = \langle\Psi|\sum_\mu Q_\mu^+ Q_\mu|\Psi\rangle. \tag{11.11}$$

The quadrupole force is attractive, therefore the system wants to increase ΔQ^2 as far as possible in the spherical single-particle potential V_0.

On the other hand, there is direct experimental information on the value of ΔQ^2 in realistic nuclei, because it is proportional to the non-energy weighted isoscalar quadrupole sum rule [see Eq. (1.43)].

$$\Delta Q^2 = \sum_\nu |\langle\Psi_\nu\|Q\|\Psi\rangle|^2 = 5\cdot\sum_\nu BE2(2_\nu^+\to 0). \tag{11.12}$$

In contrast to the energy weighted sum rule [Eq. (8.159)], which is proportional to $\langle r^2\rangle$ and is roughly a constant, the non-energy weighted sum increases dramatically during the transition from spherical to deformed: Whereas the contribution from the giant quadrupole resonance stays roughly constant, the $BE2$-value of the lowest 2^+ state increases from $\sim$20 single-particle units [see Eq. (B.85)] in the spherical case to $\sim$100 in the deformed case, where the lowest 2^+ belongs to the ground state rotational band.

The quadrupole strength, which in the spherical case is concentrated in the giant resonance, increases in the deformed case and is then concentrated into two regions, the giant quadrupole resonance and the collective rotation. In the classical limit, therefore, we have, for the spherical case only vibrations, and in the deformed case vibrations as well as rotations.

Further experimental evidence for a phase transition is the existence of a "*rotational*" *spectrum*. In Section 11.4 we shall see that in the case of strong symmetry violations we gain, besides the deformed ground state, a whole band of excited states which differ from the ground state only in the quantum number of the symmetry operator S. There, excitation energies are proportional to the eigenvalue of S^2. The simplest examples are rotational bands in deformed nuclei ($\sim I\cdot(I+1)/2\vartheta$), pairing rotations ($\sim(N-N_0)^2/8\vartheta_N$ [Sch 71a, BB 76]), or translational excitations ($\sim\mathbf{P}^2/2M$).

Phase transitions can also be described in the *boson picture*. On the one hand, we saw in Fig. 9.5 that we can, in principle, reproduce the transition from spherical to deformed shapes by a boson expansion of the fermion Hamiltonian. Most of these calculations have been carried out in a

spherical boson basis. To represent a deformed state in this basis requires very complicated wave functions, because we always work in the laboratory frame and do not violate rotational symmetry. On the other hand, we can try to represent the deformed wave function in the intrinsic frame by bosons: According to Thouless' theorem (E.26), we can represent a deformed Slater determinant in the following way.

$$|\Phi_0\rangle = \mathfrak{N} \exp\Big(\sum_{mi} Z_{mi} c_m^+ c_i \Big) |0\rangle, \tag{11.13}$$

where $|0\rangle$ is the spherical HF solution and $\mathfrak{N}$ is a normalization constant. In a well-deformed case, we have many non-vanishing coefficients, that is, the collective fermion pair operator

$$B_0^+ = \frac{1}{c} \sum_{mi} Z_{mi} c_m^+ c_i, \tag{11.14}$$

with

$$c^2 = \sum_{mi} |Z_{mi}|^2,$$

is to a good approximation a boson (see Sec. 9.2), and the deformed state

$$|\Phi) = \mathfrak{N} \exp(c B_0^+)|0)$$

is a coherent state (see Sec. 10.6.1) containing only the collective boson B_0^+. Therefore, we call it a *condensate* of the bosons B_0^+ in analogy with the Bose–Einstein condensation describing certain classes of phase transition of condensed matter (see also [SP 77a]).

This interpretation of the deformed state again shows the *classical behavior* of the mean field approximation. The structure of the boson B_0^+ is determined by the solution of the variational equations (11.2), which, in general, changes with deformation. We can show, however, that in the limit of small deformations it corresponds exactly to the *ph* part of the RPA boson, which goes to zero frequency at the transition point. In that sense, we can say that during the phase transition one boson goes to zero energy and forms a condensate. We have already used this picture in the BCS-model, where we interpreted the BCS ground state as a condensate of Cooper pairs.

So far, our discussion has concentrated mainly on the violation of rotational symmetry. In the following we show that the same considerations can be applied to symmetry violation in connection with the *pairing correlations*. Here we investigate first the simple seniority model (see Sec. 6.2). The effective Hamiltonian in a single j-shell is given by

$$H = -GS_+S_- = -G(\mathbf{S}\cdot\mathbf{S} - S_z^2 + S_z), \tag{11.15}$$

where S_i are the quasi-spin operators defined in Eq. (6.6). They act in a three-dimensional quasi-spin space. $\mathbf{S}^2$ and S_z commute with the Hamiltonian. We can therefore classify the eigenstates according to their senior-

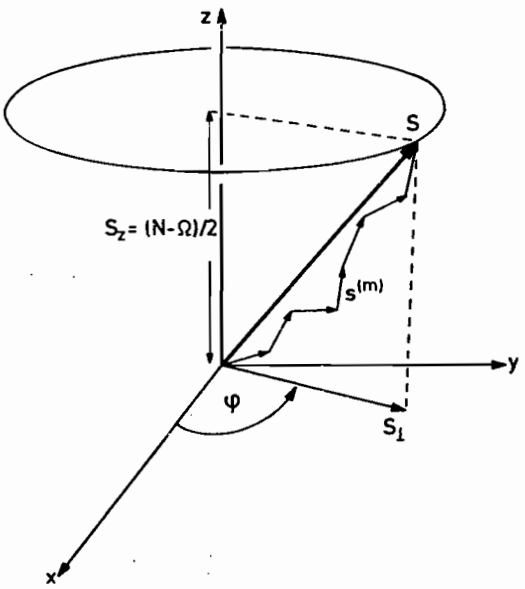


Figure 11.3. Schematic representation of the quasi-spin model.

ity s [Eq. (6.11)] and the particle number N

$$s = \Omega - 2S; \qquad \hat{N} = 2S_z + \Omega \tag{11.16}$$

S is composed by Ω quasi-spins $\mathbf{s}^{(m)}$ [see Fig. 11.3 and Eq. (6.6)]. The length of this vector is determined by the seniority s, and we will restrict the following considerations to a fixed value of s.

The other quantum number S_z corresponds to the symmetry of rotation about the z-axis. In the following we will investigate this invariance of the Hamiltonian in more detail:

$$[H, \hat{N}] = 0. \tag{11.17}$$

The particle number operator $\hat{N}$ is related to S_z (11.16) and therefore corresponds to an infinitesimal generator of rotations around the z-axis in the quasi-spin space. Such rotations are called *gauge transformations*. The corresponding angle φ is the *gauge angle*.

We can represent the operator $\hat{N}$ by this gauge angle,

$$\frac{\hat{N}}{2} = -\frac{1}{i}\frac{\partial}{\partial \varphi} + \text{const.}, \tag{11.18}$$

in analogy to the representation of the angular momentum operators **J** by Euler angles, as in Eq. (A.11).

For fixed values of N, that is, for all exact eigenstates $|\Psi\rangle$ of the Hamiltonian (11.15), the vector **S** performs a precession around the symmetry axis with a frequency

$$\frac{d\varphi}{dt} = 2\frac{dE}{dN} = 2\lambda, \tag{11.19}$$

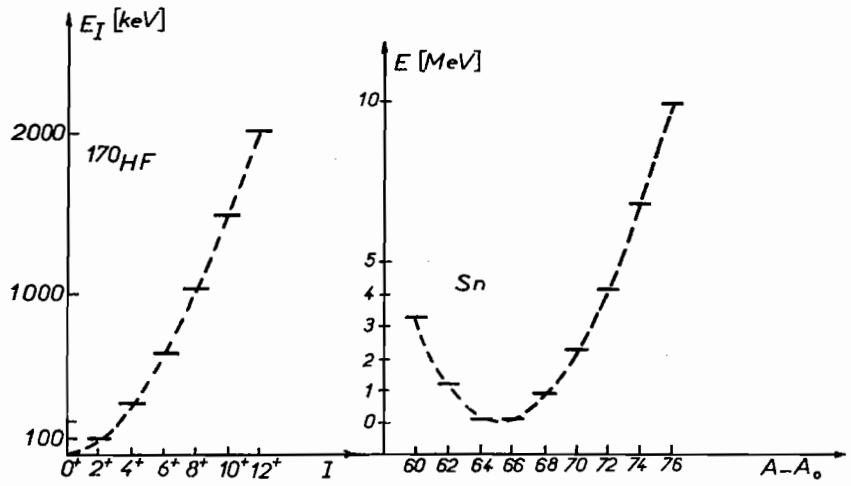


Figure 11.4. An experimental rotational spectrum (^{170}Hf), where the excited levels correspond to states with different I-values and a pairing rotational spectrum (Sn), where the "excited" levels correspond to states with different A-values (i.e., the ground states in the neighboring nuclei). (From [BB 76].)

where λ is the chemical potential [Eq. (6.41)]. We call this motion a *pairing rotation*. For different values of the particle number, we obtain a rotational band (see Fig. 11.4).

The expectation values of S_x and S_y vanish,

$$\langle\Psi|S_+|\Psi\rangle=0, \tag{11.20}$$

in the same way as the quadrupole moment of the exact ground state with $I=0$ in the case discussed above.

The length of the projection of $\mathbf{S}$ onto the x, y plane is given by

$$S_\perp^2=\langle\Psi|S_+S_-|\Psi\rangle \tag{11.21}$$

and is proportional to the correlation energy. Since there is now no "spherical" single-particle part, the system always prefers the largest possible value of $S_\perp$ in its ground state, in which all the quasi-spins $\mathbf{s}^{(m)}$ are aligned. In this model, therefore, we have for all values of the force constant a "deformed" state. For large values of S, that is, for large Ω-values and $N\ll\Omega$, we can therefore expect to obtain a good approximation to the exact wave function by breaking the gauge symmetry. The symmetry-violating inner state $|\Phi\rangle$ is a BCS wave function. It is characterized by a fixed angle φ.

$$|\Phi(\varphi)\rangle=\prod_{m>0}\left(u+ve^{i\varphi}c_m^+c_{-m}^+\right)|-\rangle=e^{i\varphi\hat{N}/2}|\Phi(0)\rangle. \tag{11.22}$$

The usual ansatz (6.31) corresponds to $\varphi=0$, that is, we have fixed the angle in a classical way. Again, all the states $|\Phi(\varphi)\rangle$ are degenerate in energy. The symmetry violation is measured by the gap parameter Δ,

which corresponds to the deformation Eq. (11.20) in the rotational case

$$\Delta = G\langle\Phi(0)|S_+|\Phi(0)\rangle = G\sum_{m>0} uv = G\cdot\Omega uv. \tag{11.23}$$

The coefficients u, v are determined by the variational principle, which yields the usual gap equation [(6.60) and (6.65)]

$$u = v = \frac{1}{\sqrt{2}}. \tag{11.24}$$

The BCS state can therefore also be represented in the quasi-boson approximation by a coherent state with quasi-boson S_+

$$|\mathrm{BCS}) = \mathfrak{N}\exp(S_+)|-). \tag{11.25}$$

It violates the particle number, but its component with the proper particle number N corresponds to the exact eigenstate (6.19) of the system. The relative fluctuations $\Delta N/N$ (6.63) vanish for large particle numbers. In this case, the mean field approximation is therefore excellent.

In more realistic applications of the BCS theory, we have the model of a pure pairing force in addition to a single-particle part

$$H = \sum_k \epsilon_k c_k^+ c_k - G\cdot P^+ P, \tag{11.26}$$

$$P^+ = \sum_{k>0} c_k^+ c_{\bar{k}}^+ .$$

An exact analytic solution is now no longer possible [KLM 61, RS 64, Ri 66]. It can be solved with the help of a computer, and the solution shows that the BCS approach is still a reasonable approximation, particularly if we use only the component with proper particle number (see Sec. 11.4). The particle number $\hat{N}$ commutes with the Hamiltonian; however, the corresponding gauge symmetry is broken by the BCS approximation and we find a phase transition. In all details the model works in full analogy to the quadrupole case (11.3). The only difference is that we now have to use a *constraint on the particle number* (by adding a term $-\lambda\hat{N}$ to the Hamiltonian), because we are not interested in a state with a vanishing expectation value of $\hat{N}$. In the case of rotational symmetry we also sometimes want wave functions with non-vanishing expectation values of J_x. We then end up with the Cranking model (see Sec. 3.4).

To show the close analogy between the transition from spherical to deformed and the transition from normal to superfluid systems, we give the following list of the quantities which correspond to one another [Sch 71a, BHR 73].

The model Hamiltonian in a spherical basis is

$$H_Q = \epsilon - \frac{\chi}{2} Q\cdot Q, \qquad H_p = \epsilon - GP^+P.$$

For small coupling strength the ground state is the uncorrelated spherical shell model state $|0\rangle$. The residual interaction, taken into account within the RPA, yields collective vibrations, namely

quadrupole vibrations with $I=2$	pairing vibrations in the nuclei with mass $A \pm 2$.

They have enhanced

$BE2$-value (in single-particle units)	pair transfer cross section (in $2p$ units)
$BE(2^+ \to 0^+) \simeq 20 B_{sp}$;	$\sigma(g.s.(A-2) \to g.s.(A)) \simeq 10 \sigma_{sp}$.

With increasing coupling strength, the lowest RPA frequency goes to zero. From this point on there exists a

deformed	superfluid

minimum in the energy surface. It is described by a product wave function based on the spherical (normal fluid) ground state $|0\rangle$

$$|\Phi\rangle = \prod_k \alpha_k |0\rangle$$

with the quasi-particle operators

$$\alpha_m^+ = \sum_k D_{km} c_k^+, \qquad \alpha_m^+ = u_m c_m^+ - v_m c_{\bar{m}},$$

$$\alpha_i^+ = \sum_k D_{ki}^* c_k, \qquad \alpha_i^+ = v_i c_i + u_i c_{\bar{i}}^+ .$$

The coefficients D_{kl} and u_k, v_k are determined from the diagonalization of the single-particle operator

$$\epsilon - q(Q + Q^+), \qquad \epsilon - \Delta(P + P^+).$$

The strength of the single-particle field is the order parameter. It is determined self-consistently by the "deformation equation" and the gap equation

$$q = \chi \langle \Phi | Q | \Phi \rangle, \qquad \Delta = G \langle \Phi | P | \Phi \rangle.$$

In the quasi-boson approximation, the wave function $|\Phi)$ can be written as a coherent state (boson condensate)

$$|\Phi) \propto \exp(c B_0^+) |0), \qquad |\Phi) \propto \exp(c A_0^+) |0),$$

with the boson

$$B_0^+ = \frac{1}{c} \sum_{mi} Z_{mi} c_m^+ c_i, \qquad A_0^+ = \frac{1}{c} \sum_m \frac{v_m}{u_m} c_m^+ c_{\bar{m}}^+ + \frac{1}{c} \sum_i \frac{u_i}{v_i} c_i c_{\bar{i}} .$$

The mean field approximation violates the symmetry.

$$[H, \hat{J}] = 0, \qquad [H, \hat{N}] = 0.$$

Its solutions $|\Phi\rangle$ characterize an orientation given by angles $\Omega = (\alpha, \beta, \gamma)$ in coordinate space, and φ in the quasi-spin space. The solutions are degenerate with respect to rotations

$$R(\Omega) = e^{i\alpha \hat{J}_z} e^{i\beta \hat{J}_y} e^{i\gamma \hat{J}_z}, \qquad G(\varphi) = e^{i\varphi \hat{N}/2}.$$

In the deformed state, we have in the excitation spectrum

β, γ vibrations (time-dependent deformation)	pairing vibrations (time-dependent gap Δ)

and (as zero-frequency modes in the RPA, see Fig. 11.4)

rotations (states with $I \neq 0$)	pairing rotations (ground states in the nuclei $A \pm 2, A \pm 4, \ldots$).

The transitions among the members of the ground state "rotational" bands are proportional to the square of the deformation parameter (q or Δ). Typical values are

$$B(E2, 2^+ \to 0^+) \simeq 100\, B_{sp}, \qquad \sigma(g.s.(A-2) \to g.s.(A)) \simeq 50 \sigma_{2p},$$

$$B(E2, 2'^+ \to 0^+) \simeq 3\, B_{sp}, \qquad \sigma(g.s.(A-2) \to p.v.(A)) \simeq 1 \sigma_{2p}.$$

The rotations can be treated (see Sec. 11.4) either classically by the Cranking model:

$$H - \omega \hat{J}_x, \qquad H - \lambda \hat{N};$$

or quantum mechanically by projection

$$P^I_{MK} \propto \int D^{I*}_{MK}(\Omega) R(\Omega)\, d\Omega, \qquad P^N \propto \int e^{-i(\varphi/2)N} G(\varphi)\, d\varphi.$$

This list concludes our general considerations about symmetries and its violations within the mean field theory, and it is clear that a similar list can be produced for all kind of phase transitions connected with a symmetry violation, for instance, pion condensation. We shall now turn to methods that allow a restoration of the broken symmetries.

11.3 Transformation to an Intrinsic System

11.3.1 General Concepts

Symmetry violations in the many-body wave functions are strongly connected with *collective motion*. To explain this fact in more detail we again use the example of rotational symmetry. A deformed wave function $|\Phi\rangle$ defines a fixed orientation in space, characterized, for instance, by the principal axes of the mass distribution. All wave functions $|\Phi(\Omega)\rangle$ that differ from $|\Phi\rangle$ only in this orientation have the same internal structure and yield the same energy expectation value. They lie at the bottom of a valley in the energy surface. In the classical picture we therefore have a type of motion which stays, if it is slow enough, close to the bottom of this valley. It corresponds to rotational motion with an approximate preservation of the intrinsic structure. If the deformation is large, there are many particles involved in this motion and we call it collective. (For a further discussion of these quasi-classical considerations, see also Sec. 12.3.)

In quantum mechanics, the states $|\Phi(\Omega)\rangle$ span a subspace of the many-body Hilbert space, the collective subspace in the sense of the GCM method (see Chap. 10). If this subspace is large enough—that is, as we shall see in Section 11.4, if the overlap $\langle \Phi(\Omega) | \Phi(\Omega') \rangle$ is strongly peaked at $\Omega = \Omega'$, which happens for strong deformations—it contains not only the

ground state to a good approximation, but also low-lying excited states, with the same internal structure, which exhibit a rotational spectrum.

Since the intrinsic structure in such a wave function is preserved to a good approximation in the collective motion, many authors have tried to separate intrinsic and collective degrees of freedom by an appropriate coordinate transformation from the $3A$ particle coordinates $\mathbf{r}_i$ to collective coordinates $q_1,\ldots,q_f$ and intrinsic coordinates $\xi_1 \ldots \xi_{3A-f}$ [Vi 57a and b, Vi 58, SG 68, FKS 72]

$$(\mathbf{r}_1,\ldots,\mathbf{r}_A)\rightarrow(q_1,\ldots,q_f,\xi_1,\ldots,\xi_{3A-f}). \tag{11.27}$$

The collective coordinates are usually well defined by the underlying symmetry as the canonical conjugate variables to the generators of the symmetry group. In the case of translations, there are the center of mass coordinates; in the case of rotations, there is a set of angles as the Euler angles; and in the case of superfluid systems, it is the Gauge angle φ.

The definition of the intrinsic variables ξ is a serious problem. It is only simple in the translational case of two particles, where it is the relative coordinate $\mathbf{r}=\mathbf{r}_1-\mathbf{r}_2$. In the case of a few particles, we can introduce, for instance, Jacobi coordinates, but this procedure in the translational case and similar procedures in the rotational case get extremely complicated in the case of our many-particle system. Nevertheless as we shall see in some examples, we can sometimes deduce the structure of the Hamiltonian in the collective variables without explicitly specifying the intrinsic coordinates.

The main idea of these theories is to decompose the nuclear Hamiltonian into the following three parts

$$H=H_{\text{coll}}(q)+H_{\text{intr}}(\xi)+H_{\text{coupl}}(q,\xi), \tag{11.28}$$

where H_{coll} describes the collective and H_{intr} the intrinsic motion. The coupling term H_{coupl}, it is hoped, will be small or of a simple structure. The case of nuclear translations is an example of where this concept works well. Because of Galilean invariance, H_{coupl} vanishes identically.

If we can neglect H_{coupl}, we find the eigenfunctions of the total system as products of a collective wave function $\chi_c(q)$ and an intrinsic wave function $\varphi_n(\xi)$

$$\Psi_n^c=\chi_c(q)\cdot\varphi_n(\xi), \tag{11.29}$$

and the energies have the form

$$E_n^c=E_c+E_n. \tag{11.30}$$

For each intrinsic state φ_n there exists a collective band labelled with the index c.

In practical applications of this procedure, we are always faced with two problems:

(i) The decomposition (11.28) is not unique. It depends on the actual choice of the intrinsic coordinates. It is evident that the size of the coupling term depends sensitively on this choice. Therefore, some-

times the variational principle of minimal coupling has been used to determine the intrinsic system [Co 56, BM 58, RV 69, Ru 73].

(ii) The intrinsic Hamiltonian $H_{\text{intr}}(\xi)$ does not depend on $3A$ coordinates of A particles, but only on $3A-f$ variables ξ. None of them can be identified with particle position coordinates. This is particularly unpleasant, if we want to describe the intrinsic motion within the independent particle picture. We already have this problem in the case of translations.

A series of authors have therefore used the concept of *redundant variables* [LST 55, Li 55, Ta 56b, Na 57, Li 58, VC 70]. Here it is attempted to combine the advantages of the method of canonical transformations with the advantages of dealing with $3A$-particle coordinates to describe the intrinsic motion. This is achieved by embedding the A-body Hilbert space in the particle coordinates $\mathbf{r}_1,\ldots,\mathbf{r}_A$ (the physical subspace) into a larger space with the dimension $3A+f$, where f is the number of collective variables. We characterize these additional unphysical degrees of freedom by the coordinates $g_1,\ldots,g_f$. The physical subspace is then characterized by the conditions

$$g_1=g_2=\cdots=g_f=0. \tag{11.31}$$

Physical operators, that is, operators which depend only on the coordinates $\mathbf{r}_1,\ldots,\mathbf{r}_A$ as, for instance, the Hamiltonian, have eigenfunctions of the form

$$\Psi_{mn}(g,\mathbf{r})=h_m(g)\varphi_n(\mathbf{r}), \tag{11.32}$$

where $h_m(g)$ is an arbitrary orthonormal set of functions in the redundant coordinates. Each eigenvalue E_n is therefore highly degenerate in the enlarged space, but the corresponding eigenfunctions are identical in the physical subspace (11.31).* We are now able to transform the coordinates $g_1\ldots g_f, \mathbf{r}_1\ldots\mathbf{r}_A$ in the redundant laboratory system into the redundant intrinsic system with the collective coordinates $q_1,\ldots,q_f$ and the particle coordinates $\mathbf{x}_1,\ldots,\mathbf{x}_A$:

$$\begin{aligned}\mathbf{r}_i&=\mathbf{r}_i(q_1,\ldots,q_f,\mathbf{x}_1,\ldots,\mathbf{x}_A), \qquad i=1,\ldots,A\\ g_i&=g_i(q_1,\ldots,q_f,\mathbf{x}_1,\ldots,\mathbf{x}_A), \qquad i=1,\ldots,f.\end{aligned} \tag{11.33}$$

In this way we derive a Hamiltonian in the variables $\mathbf{q}$ and $\mathbf{x}$ of the form

$$H=H_{\text{coll}}(\mathbf{q})+H_{\text{intr}}(\mathbf{x})+H_{\text{coupl}}(\mathbf{q},\mathbf{x}). \tag{11.34}$$

In the next step we apply the usual approximation techniques for the diagonalization of H_{intr}. The question remains, however, as to whether we can find an appropriate transformation (11.33) such that the coupling term is small, and also whether we can separate the admixture of spurious states obtained by such approximations in a proper way. Before we discuss these techniques in the context of translations and rotations, we wish to mention

* Further details concerning the calculation with redundant variables are given in the quantum mechanical case by Lipkin et al. [LST 55], and for the classical theory by Watanabe [Wa 56].

that similar methods have also been applied for more general collective motions such as surface vibrations, which are not connected with a broken symmetry [BP 53, Sü 54, MYT 55, To 55, MY 55, Co 55, MT 56, HM 57, Vi 57b], but for which one is also interested in deriving the collective Hamiltonian.

11.3.2 Translational Motion

In the following we will study translational motion in more detail. This simple case already shows many problems. In nuclear physics it is connected with a very strong symmetry violation. It is the only case where the collective motion decouples completely from the intrinsic degrees of freedom. The reason for this is Galilean invariance.

The intrinsic system in this case is given by the center of mass frame, and the collective coordinate is

$$\mathbf{R}=\frac{1}{A}\sum_i \mathbf{r}_i. \tag{11.35}$$

In a first step we do not use redundant coordinates. Without specifying the intrinsic coordinates ξ_l we gain from (11.27) for the derivatives

$$\frac{\partial}{\partial \mathbf{r}_i}=\frac{1}{A}\frac{\partial}{\partial \mathbf{R}}+\sum_l \frac{\partial \xi_l}{\partial \mathbf{r}_i}\frac{\partial}{\partial \xi_l}. \tag{11.36}$$

Since the intrinsic coordinates ξ_l depend only on the differences $\mathbf{r}_i-\mathbf{r}_j$, we get $\sum_i(\partial/\partial \mathbf{r}_i)\xi_l=0$ and find a complete decoupling of the collective motion

$$H=-\frac{\hbar^2}{2m}\sum_i\left(\frac{\partial}{\partial \mathbf{r}_i}\right)^2+V=\frac{1}{2Am}\mathbf{P}^2+\sum_{ll'}D_{ll'}(\xi)\pi_l\pi_{l'}+V(\xi), \tag{11.37}$$

with the center of mass momentum $\mathbf{P}=(\hbar/i)\partial/\partial \mathbf{R}$, the intrinsic momenta $\pi_l=(\hbar/i)\partial/\partial\xi_l$, and the intrinsic inertia functions

$$D_{ll'}(\xi)=\sum_i \frac{\partial \xi_l}{\partial \mathbf{r}_i}\cdot\frac{\partial \xi_{l'}}{\partial \mathbf{r}_i}. \tag{11.38}$$

As discussed in the last section, we cannot approximate the intrinsic part of (11.37) by a shell model Hamiltonian. Therefore, we now apply the method of redundant coordinates. The coordinates $\mathbf{x}_i$ are in this case the particle coordinates in the center of mass frame, and for the transformation (11.33) we get

$$\begin{aligned} &\mathbf{r}_i=\mathbf{x}_i+\mathbf{R} && \mathbf{x}_i=\mathbf{r}_i-\frac{1}{A}\sum_j \mathbf{r}_j+\mathbf{g} \\ &\mathbf{g}=\frac{1}{A}\sum_j \mathbf{x}_j && \mathbf{R}=\frac{1}{A}\sum_j \mathbf{r}_j-\mathbf{g}. \end{aligned} \qquad \text{or} \tag{11.39}$$

The physical subspace is characterized by $\mathbf{g}=0$. For the momenta we find the transformation

$$\frac{\hbar}{i}\frac{\partial}{\partial \mathbf{r}_i}=\frac{1}{A}\mathbf{P}+\mathbf{p}_i-\frac{1}{A}\sum_j \mathbf{p}_j, \tag{11.40}$$

with

$$\mathbf{p}_i=\frac{\hbar}{i}\frac{\partial}{\partial \mathbf{x}_i}, \qquad \mathbf{P}=\frac{\hbar}{i}\frac{\partial}{\partial \mathbf{R}}.$$

The Hamiltonian in the coordinates $\mathbf{x}$, $\mathbf{R}$ takes the form

$$H=\frac{1}{2Am}\mathbf{P}^2+\sum_i\frac{\mathbf{p}_i^2}{2m}+V(\mathbf{x}_i-\mathbf{x}_j)-\frac{1}{2Am}\left(\sum_i\mathbf{p}_i\right)^2. \tag{11.41}$$

Again we get a complete decoupling. The collective Hamiltonian $(1/2Am)\mathbf{P}^2$ has plane waves as eigenfunctions. To each internal eigenstate exists a continuum corresponding to the translational motion. Since we are only interested in intrinsic excitations, it is enough to investigate only H_{intr}:

$$H_{\text{intr}}=\sum_i\frac{\mathbf{p}_i^2}{2m}+V-\frac{1}{2Am}\left(\sum_i\mathbf{p}_i\right)^2. \tag{11.42}$$

It looks very similar to the original Hamiltonian H in the coordinates $\mathbf{r}_i$. Only the term $(1/2Am)\sum_{ij}\mathbf{p}_i\mathbf{p}_j$, which is sometimes called the "recoil term," is subtracted. We are now able to treat this intrinsic Hamiltonian with the usual techniques. We do not have to worry about translational invariance, because we are in the intrinsic system. Using, for instance, the Hartree–Fock method, we would get a localized potential and localized wave functions. There are only two things we have to take into account:

(i) We have to subtract from the usual Hamiltonian the term $(1/2Am)\sum_{ij}\mathbf{p}_i\mathbf{p}_j$, which contains a two-body interaction. This is often done in HF calculations. In a pure phenomenological shell model, we sometimes subtract at least the expectation value of this operator from the final energy. It is the kinetic energy of the spurious center of mass motion.*

(ii) We have to be aware that because we use redundant coordinates there are spurious solutions among our eigenfunctions. In cases in which we solve the problem in the center of mass frame exactly they do not mix with the physical solutions and can be separated out by special techniques. In the case of an approximate solution, however, they can mix with physical solutions and it is sometimes hard to decide which states are physical and which are not.

To understand the structure of these unphysical solutions a little better, we use a purely harmonic interaction [GS 57]

$$V=\frac{k}{2}\sum_{i<j}(\mathbf{r}_i-\mathbf{r}_j)^2. \tag{11.43}$$

In this case we can write

$$H_{\text{intr}}=H_0+H_1, \tag{11.44}$$

with

$$H_0=\sum_i\left(\frac{\mathbf{p}_i^2}{2m}+\frac{Ak}{2}\mathbf{x}_i^2\right) \tag{11.45}$$

and

$$H_1=-\frac{1}{2Am}\left(\sum_i\mathbf{p}_i\right)^2-\frac{A^2k}{2}\left(\frac{1}{A}\sum_i\mathbf{x}_i\right)^2. \tag{11.46}$$

*The same term is automatically obtained, if we work with the Hamiltonian in the laboratory system (coordinates $\mathbf{r}_i$) and treat the ground state correlations in the framework of the RPA approach [see Eq. (8.111)]. It is the contribution of the zero-frequency mode connected with the translation to the ground state correlation energy. The RPA is a symmetry conserving theory. It is therefore not necessary to introduce redundant coordinates here. We also get the same term in other symmetry-conserving theories, like projection techniques [see Sec. (11.4.4)].

H_0 is a single-particle harmonic oscillator potential with corresponding Slater determinants as eigenfunctions. Since H_1 commutes with H_0 it can be simultaneously diagonalized, and the exact solution of H_{intr} is just the shell model wave function corrected for the center of mass motion. On the other hand, H_1 describes an oscillator of the center of mass $\mathbf{g}=(1/A)\sum_i \mathbf{x}_i$ of the intrinsic coordinates in an harmonic oscillator potential with the frequency $\omega=(Ak/m)^{1/2}$ of an individual particle. This motion is certainly spurious, because in the physical space we have the condition $\mathbf{g}=0$. The spurious solutions correspond to oscillations of the nucleus as a whole in the shell model potential.

In shell model calculations based on an harmonic oscillator basis they can be separated by group theoretical methods [BR 37, ES 55, UT 58, Kr 60, Ve 60, KM 66, AMK 69, Mo 69, He 71, Sch 77b]. Since the center of mass problems are especially important in light nuclei, and since, on the other hand, the harmonic oscillator potential is a reasonable approximation for the self-consistent field in this region of the periodic table, this basis is preferred for many practical applications.

In the general case the problem is very complex. An exact separation of the spurious components is often hardly possible. It has been tried to remove the spurious components approximatively by adding an operator $S(\mathbf{g})$ to the Hamiltonian, which vanishes in the physical space ($\mathbf{g}=0$) and is large and positive for $\mathbf{g}\neq 0$. It shifts the spurious eigenstates to high energies such that they do not mix with low-lying states even if they are only calculated approximately [Pa 67, SG 68, FKS 72].

In heavy nuclei the spurious center of mass motion can be neglected in all cases, where it gives an effect only of the order $1/A$. As an example, we calculate the orbital part of the magnetic moment in a nucleus with one odd nucleon outside a magic configuration (for instance, Pb^{209} or Bi^{209}). The orbital angular momentum of the ith particle with respect to the center of mass of the whole system is given by

$$\mathbf{l}_i=\mathbf{x}_i\times\mathbf{p}_i=\left(\mathbf{x}_i'-\frac{1}{A}\mathbf{x}_0'\right)\times\left(\mathbf{p}_i'-\frac{1}{A}\mathbf{p}_0'\right), \tag{11.47}$$

where x_i', p_i' are the coordinates of the particles in the center of mass system of the magic core. The index 0 characterizes the odd particle. Only the protons contribute to the orbital part of the magnetic moment. If p characterizes the protons in the core, we get

$$\begin{aligned}\frac{\mu}{\mu_k}&=\sum_p \mathbf{l}_p+g_l\cdot\mathbf{l}_0\\&=\sum_p \mathbf{l}_p'-\frac{1}{A}\left(\sum_p \mathbf{x}_p'\right)\times\mathbf{p}_0'-\frac{1}{A}\mathbf{x}_0'\times\left(\sum_p \mathbf{p}_p'\right)+\left(\frac{Z}{A^2}+g_l\left(1-\frac{2}{A}\right)\right)\mathbf{l}_0'.\end{aligned} \tag{11.48}$$

Depending on whether the odd particle is a proton or a neutron, g_l is one or zero, respectively. Within a shell model approach whose center coincides with the center of mass of the magic core, all the protons are in closed shells and the first three terms in Eq. (11.48) vanish and we are left with a g-factor corrected for the center of mass motion $\tilde{g}_l$:

$$\tilde{g}_l=g_l\left(1-\frac{2}{A}\right)+\frac{Z}{A^2}. \tag{11.49}$$

This is a $1/A$ effect.

11.3.3 Rotational Motion

The rotational case has many formal similarities to the translational case; there are, however, some basic differences:

(i) There is no counterpart to the Galilean invariance in the rotational case. Therefore, we cannot expect a priori decoupling between collective and intrinsic motion. We only can hope that the coupling terms become small at least for not-too-large angular momenta. How large they are depends on the choice of the intrinsic system.

(ii) There is no well-defined center of mass angle. The most natural choice of the intrinsic system seems to be provided by the principal axis of the mass distribution. It turns out, however, that the spurious motion of the system around these axes yields very large coupling terms between intrinsic and collective degrees of freedom [VC 70].

(iii) The underlying symmetry group, the rotational group, is no longer Abelian. Whereas the first two difficulties are already encountered in the one-dimensional rotations in a plane,* we have additional difficulties with non-commuting operators. This requires some lengthy calculations involving angular momentum algebra, which we do not want to go into here. The reader is referred to the literature for all the details [LST 55, Na 57, Vi 57a, b]. We shall present only the main results.

As collective coordinates, Euler angles $\Omega=(\alpha, \beta, \gamma)$ are chosen that connect the laboratory frame to the body-fixed system, whose precise definition is still open. In a first step, we can again try to use $3A-3$ intrinsic coordinates ξ. Villars [Vi 57a, b] in this case derived the following general structure of the Hamiltonian

$$H=\frac{1}{2}\sum_{ik=1}^{3}\left(\vartheta^{-1}(\xi)\right)_{ik}\hat{I}_i\cdot\hat{I}_k+\sum_{i=1}^{3}B_i(\xi)\hat{I}_i+H_{\text{intr}}(\xi). \tag{11.50}$$

The operators $\hat{I}_i$ are the components of the angular momentum operator $\mathbf{I}$ with respect to the intrinsic axis $i=(1, 2, 3)$ expressed by Euler angles [see Eq. (A.14)]. The rest of the Hamiltonian, particularly the tensor of inertia ϑ_{ik} and the quantities B_i, depend only on the intrinsic coordinates ξ. They provide a coupling between the collective and the intrinsic motion.

If we determine the body-fixed system by the principal axis of the mass distribution, we find that the tensor ϑ_{ik} is diagonal and that the moments of inertia are given by the hydrodynamical values (1.48) derived from an irrotational liquid drop motion. They are in strong disagreement with experimental evidence. Consequently, the coupling terms B_i have to be very large.

We can also give the general structure of the eigenfunctions to the Hamiltonian (11.50). Since it is rotational invariant, and since the Wigner functions $D^{I*}_{MK}(\Omega)$ provide a complete system in the variables Ω, we find for the eigenfunctions of Eq. (11.50):

$$\Psi^{IM}(\Omega, \xi)=\sum_{K}D^{I*}_{MK}(\Omega)\cdot g_K(\xi). \tag{11.51}$$

As in the translational case, the coordinates ξ are not very useful for practical

*The violation of gauge symmetry in the BCS wave function would correspond to such a one-dimensional rotation.

applications. Several authors therefore used the concept of redundant coordinates. Again we are faced with the problem of a proper choice of the body-fixed frame. If we choose the principal axes of the mass distribution, we again find the irrotational moments of inertia and large coupling terms. Several attempts have been made to remove them by a unitary transformation or a proper choice of the intrinsic system [Co 56, HM 57, BM 58, VC 70, Ru 73]. However, in general, in this way we end up with very complicated equations. Their solution is only possible with more or less drastic approximations. The most elaborate investigations of this kind have been carried out by Villars and Cooper [VC 70]. They obtain a renormalization of the irrotational moment of inertia and end up with the Thouless–Valatin value [Eq. (8.113)].

Summarizing all these investigations, we can say that the transformation to the intrinsic system and the construction of a collective Hamiltonian in this way encounters serious problems that have not yet been solved completely. It is only in the case of translations that complete separation of the collective and intrinsic degrees of freedom can be achieved, and where in the case of an harmonic oscillator basis an elimination of spurious results is possible by group-theoretical methods.

In view of the great success of the rotor model of Bohr and Mottelson (see Chaps. 1 and 3), however, we should expect that it is also possible in the rotational case to achieve at least an approximate decoupling of collective and intrinsic motion. The essential problem is the determination of the moments of inertia. In contradiction to the translational case, where the total mass is independent of the residual interaction, the inertia parameters of the rotational motion are influenced by the correlations. This is a dynamic effect which can hardly be taken into account by a pure coordinate transformation.

A much more general way to treat symmetry violations is given by the method of generator coordinates. In many cases these techniques correspond to a projection onto the corresponding eigenspace of the symmetry operator. We will treat these projection methods in the next section.

11.4 Projection Methods

11.4.1 Projection Operators

Projection techniques are a special case of the generator coordinate method (GCM) discussed in Chapter 10. The basic concept of this method is a diagonalization of the full many-body Hamiltonian H in the subspace spanned by a set of generating functions $|\Phi(\Omega)\rangle$, which depend on one or several continuous parameters Ω. This space is called the "collective subspace."

We now have a symmetry group, generated by the symmetry operators S_i, which commute with the Hamiltonian, and a symmetry-violating wave function* $|\Phi\rangle$. Applying all the elements $R(\Omega)$ of this group onto $|\Phi\rangle$, we

*In the following we always have product wave functions in mind, although the method can be generalized to other types of symmetry-breaking functions.

get a continuous set of generating functions

$$|\Phi(\Omega)\rangle = R(\Omega)|\Phi\rangle, \tag{11.52}$$

where Ω runs over the whole parameter space of the group.

All the wave functions $|\Phi(\Omega)\rangle$ have the same energy expectation value. Peierls and Yoccoz [PY 57] therefore proposed to use them as generating functions in the GCM technique, that is, to make the following ansatz for the wave function $|\Psi\rangle$ of the system*

$$|\Psi\rangle = \int d\Omega\, f(\Omega)|\Phi(\Omega)\rangle. \tag{11.53}$$

The variation of the energy with respect to the weight function $f(\Omega)$ corresponds to a diagonalization of H in the space spanned by the functions $|\Phi(\Omega)\rangle$ (see Sec. 10.2). This subspace is invariant under the elements of the symmetry group [Ze 67]:†

$$R(\Omega)|\Psi\rangle = \int d\Omega'\, f(-\Omega+\Omega')|\Phi(\Omega')\rangle, \tag{11.54}$$

which means that the projector P_Φ onto this subspace commutes with the symmetry operation $R(\Omega)$. Thus we see that we may find simultaneous eigenfunctions of $P_\Phi H P_\Phi$ and the symmetry operators. *In other words, there exists a function $f(\Omega)$ which minimizes the energy and causes $|\Psi\rangle$ to have the proper symmetry.*

To find such a function, we expand $f(\Omega)$ in a complete set of eigenfunctions of the symmetry operators expressed in the variables Ω. For all Abelian groups this corresponds to a Fourier decomposition. As an example, we use the case of the gauge group connected with particle number violation (see Sec. 11.2). It has one group parameter φ, and the symmetry operator $\hat{N}/2$ has the representation $i\partial/\partial\varphi$. Its eigenfunctions are exponentials

$$f(\varphi) = \sum_n \frac{1}{2\pi} e^{-in\varphi} \cdot g_n \tag{11.55}$$

with integer n because of the periodicity in φ. From Eqs. (11.53) and (11.55) we get

$$|\Psi\rangle = \sum_n g_n \hat{P}^{2n}|\Phi\rangle, \tag{11.56}$$

with

$$\hat{P}^A = \frac{1}{2\pi}\int_0^{2\pi} e^{i\varphi(\hat{N}-A)}\, d\varphi. \tag{11.57}$$

The operator $\hat{P}^A$ is a projection operator in the mathematical sense ($\hat{P}^2 = \hat{P}, \hat{P}^+ = \hat{P}$), which projects onto the subspace with particle number

*For a review of projection methods based on the GCM ansatz, see also [Wo 75] and, for a review on more general projectors, see [Ma 70].

†Note that for non Abelian groups $-\Omega+\Omega'$ depends on the ordering and means the set of parameters corresponding to the transformation $R^{-1}(\Omega)R(\Omega')$.

A. To see this, we multiply $\hat{P}^A$ from the left and from the right with the unity operator, expressed by a complete orthogonal set of wave functions $|N\alpha\rangle$ with the particle number N:

$$\hat{P}^A = \sum_{N\alpha,\, N'\alpha'} |N\alpha\rangle\langle N\alpha| \frac{1}{2\pi} \int_0^{2\pi} e^{i\varphi(\hat{N}-A)} |N'\alpha'\rangle\langle N'\alpha'|$$

$$= \sum_{N\alpha,\, N'\alpha'} |N\alpha\rangle\langle N\alpha|N'\alpha'\rangle\langle N'\alpha'| \cdot \delta_{NA} = \sum_{\alpha} |A\alpha\rangle\langle A\alpha|. \quad (11.58)$$

Since we are only interested in a wave function $|\Psi\rangle$ with the proper particle number A, we find

$$g_n = 0 \quad \text{for} \quad 2n \neq A.$$

$g_{A/2}$ is a normalization constant.

In the same way, in the case of translations we obtain a projector onto eigenstates of the linear momentum operators $\mathbf{P}$ with eigenvalues $\mathbf{p}$:

$$\hat{P}^{\mathbf{p}} = \frac{1}{(2\pi\hbar)^3} \int_{-\infty}^{+\infty} e^{(i/\hbar)\mathbf{q}(\mathbf{P}-\mathbf{p})}\, d^3q = \delta(\mathbf{P}-\mathbf{p}) \quad (11.59)$$

and

$$\hat{P}^{\mathbf{p}}\Phi(\mathbf{r}_1,\ldots,\mathbf{r}_A) = \int d^3q\, e^{i\mathbf{p}\mathbf{q}/\hbar}\Phi(\mathbf{r}_1-\mathbf{q},\ldots,\mathbf{r}_A-\mathbf{q}).$$

For Abelian symmetry groups the weight function f is therefore completely determined by pure kinematics.

This is different for non-Abelian groups. Here we do not end up with a projector in the mathematical sense—though we also use the term "projection techniques" in this case. The weight function f is not completely determined by the symmetry. The most important example is the case of rotations in three dimensions. Since this symmetry group involves many details, and we do not want to interrupt the general discussion of projection techniques here, we will come back to three-dimensional rotations in Section 11.4.6.

11.4.2 Projection Before and After the Variation

So far we have not specified the symmetry-violating internal wave function $|\Phi\rangle$. In principle, there are two ways to determine this function

(i) *Variation before the projection* (VBP) [PY 57]. In this case we proceed, for instance, as described in Chapter 7 and determine the optimal product wave function $|\Phi\rangle$ by the variational principle

$$\delta \frac{\langle\Phi|H|\Phi\rangle}{\langle\Phi|\Phi\rangle} = 0; \quad (11.60)$$

A "deformed" solution $|\Phi\rangle$ of this equation is a superposition of eigenstates of the corresponding symmetry operator, for instance, of

the angular momentum* $\hat{J}$: Applying the projection operator $\hat{P}^I$, we get a wave function

$$|\Psi^I\rangle = \hat{P}^I|\Phi\rangle, \tag{11.61}$$

which contains only components with the angular momentum I. It is no longer a product wave function, but a complicated superposition of Slater-determinants as discussed in Section 5.5. Insofar it contains many more correlations than the function $|\Phi\rangle$. As we shall see in Section 11.4.4, it therefore has a lower energy expectation value and it is certainly a much better approximation of the exact wave function of the system.

The method is rather simple insofar as we get the whole rotational band based on the intrinsic wave function $|\Phi\rangle$ if we apply the projector $\hat{P}^I$ for different values of I. The variational equation (11.60) has to be solved only once. However, this method does violate the variational principle, since we do not vary the projected function $|\Psi^I\rangle$ and it does not allow for changes of the self-consistent internal field within a rotational band. We can therefore improve the method in the following way.

(ii) *Variation after projection* (VAP): [Ze 65, RY 66, Yo 66]. Here we use the proper variational principle

$$\delta \frac{\langle \Psi^I|H|\Psi^I\rangle}{\langle \Psi^I|\Psi^I\rangle} = \delta \frac{\langle \Phi|\hat{P}^I H \hat{P}^I|\Phi\rangle}{\langle \Phi|\hat{P}^I\hat{P}^I|\Phi\rangle} = 0, \tag{11.62}$$

which means that we have to minimize the expectation value of the projected energy $\hat{P}^I H \hat{P}^I$ within the set of product wave functions $|\Phi\rangle$.

In principle, this method corresponds to a double variational GCM technique [see Eq. (10.147)], where we use the ansatz

$$|\Psi\rangle = \int d\Omega\, f(\Omega) R(\Omega)|\Phi\rangle \tag{11.63}$$

and vary the energy with respect to both the weight function f and the generating function $|\Phi\rangle$. Since the function f is determined more or less by the symmetry, only a variation of $|\Phi\rangle$ is left.

This method is certainly better than a variation before projection, but is also much more complicated. We must repeat the variation for each value of I again, and the operator $\hat{P}^I H \hat{P}^I$ is a multi-body operator.

Only in the case of particle number violation within the BCS model (Chap. 6) have both methods been carried out exactly. We will discuss these calculations in more detail in Section 11.4.3.

*Since in this section we give only a very qualitative discussion of different projection techniques, we use the angular momentum $\hat{J}$ here only as an example, without going into the details of the rotational group. $\hat{J}$ could be replaced in the following by any other symmetry operator.

In the BCS case we have only to vary with respect to the occupation probabilities v_k^2. In the general case, where we must also vary the single-particle wave functions φ_k, the VAP methods become extremely complicated. In principle, we can write down (in analogy to Sec. 7.3.1) projected HFB equations in the most general case [Ze 65, On 68, SG 79], but in detail they correspond to a matrix of integral equations in several variables that have not been solved so far.

In all practical applications, therefore, one has used, to a greater or lesser extent, rather drastic approximations. There are two ways to solve Eq. (11.62) approximately:

(i) We use the exact projection operator and restrict the set of trial functions $|\Phi\rangle$ drastically, allowing only very few important degrees of freedom. For instance, we can use a set of wave functions $|\Phi(\beta,\gamma)\rangle$ depending only on some deformation parameters, which we obtain from a diagonalization of the corresponding Nilsson potential. For each point in the (β,γ) plane we then have to calculate the projected energy and search for minima in this surface. This method is the complete analogy to the technique used in Fig. 7.1 for the determination of the unprojected HF solution. It has been applied in calculations in the high spin region, where particle number projection is important [DBM 73, FLW 73, FDG 76].

(ii) We can use rather general assumptions for the wave functions $|\Phi\rangle$, such as large deformations and nearly Gaussian shape for the overlap kernels [see Eq. (11.79)] to approximate the projected energy without specifying the detailed structure of $|\Phi\rangle$. These techniques correspond to the momentum expansion in the GCM method (see Sec. 10.7). We will discuss them in Section 11.4.4.

Here we want only to anticipate the most important result [Ka 68, BMR 70, VS 71, Co 72]: In the limit of strong deformations, we find that the cranking model wave function $|\Phi_\omega\rangle$ (see Sec. 7.7) is an approximate solution of Eq. (11.62). This wave function is defined as a solution of the constrained variational principle

$$\delta\frac{\langle\Phi|H-\omega\hat{J}_x|\Phi\rangle}{\langle\Phi|\Phi\rangle}=0, \qquad \langle\Phi|\hat{J}_x|\Phi\rangle=\sqrt{I(I+1)}\,, \tag{11.64}$$

and ω is determined in such a way that the symmetry is, on the average, conserved. To each I-value we must therefore use a different internal wave function.*

*In the BCS case this result is rather trivial, because we adjust the chemical potential anyway for each value of N. This is necessary, because the simplest version of VBP without symmetry conservation on the average, would be a very poor approximation. The reason for that is given by the fact that the BCS wave function is not "strongly deformed," because BCS correlations are usually rather weak in nuclei. The first-order variation after the projection method, as it is given by the projection after variation from a BCS-function with proper particle number (PBCS), can therefore be improved by a full variation after projection (FBCS).

11.4.3 Particle Number Projection

The violation of the particle number in BCS wave functions is a nice example of a symmetry violation that can be treated by projection techniques.* Over the years a large number of methods have been developed in this field [Ba 60b, KLM 61, DMP 64, MPR 65, La 65, MRR 66, Ri 66, DK 66, IO 67, Fo 70, So 72, FHM 73, GGF 73, MMW 73, DMR 75, MR 77]. As an example, we present the *method of residues* proposed by Dietrich, Mang, and Pradal [DMP 64]. It starts with the projector (11.57). Introducing a complex variable $z = e^{i\varphi}$, we can write it as an integral in the complex plane,

$$\hat{P}^A = \frac{1}{2\pi i} \oint \frac{z^{\hat{N}}}{z^{A+1}} dz. \tag{11.65}$$

When applied to a BCS wave function $|\Phi\rangle$ of the form (6.31), we get

$$|\Psi^A\rangle = \hat{P}^{A-2p}|\Phi\rangle = \frac{1}{2\pi i} \oint \frac{d\zeta}{\zeta^{p+1}} \prod_k (u_k + v_k \zeta a_k^+ a_{\bar{k}}^+)|-\rangle, \tag{11.66}$$

where we have replaced z by $\zeta = z^2$ and used the fact that the pair of operators $a_k^+ a_{\bar{k}}^+$ raises the particle number by two. $p = A/2$ is the number of pairs. The integrand in (11.66) is a Laurent series in ζ. The integration just picks the component with ζ^{-1}, that is, the component with p pairs.

Using the fermi commutation relations for the operators a_k, a_k^+, we can express arbitrary matrix elements by the residues

$$R_\nu^m(k_1, \ldots, k_m) = \frac{1}{2\pi i} \oint \frac{dz}{z^{(p-\nu)+1}} \prod_{\substack{k \neq k_1, \ldots, k_m \\ >0}} (u_k^2 + z v_k^2). \tag{11.67}$$

The states listed in the argument of $R(\ldots)$ are to be excluded from the product under the integral. As simple examples, we find the norm

$$\langle \Psi^A | \Psi^A \rangle = R_0^0 = \frac{1}{2\pi i} \oint \frac{dz}{z^{p+1}} \prod_{k>0} (u_k^2 + z v_k^2) \tag{11.68}$$

and the occupation probabilities for the level k in the wave function $|\Psi^A\rangle$

$$\frac{\langle \Psi^A | a_k^+ a_k | \Psi^A \rangle}{\langle \Psi^A | \Psi^A \rangle} = v_k^2 \cdot \frac{R_1^1(k)}{R_0^0}, \tag{11.69}$$

which shows that in the projected BCS theory the numbers v_k^2 no longer have the meaning of occupation probabilities. They are only variational parameters.

The residues R_ν^m can be evaluated in a rather simple way from the recursion relations

$$R_\nu^m(k_1, \ldots, k_m) = u_k^2 R_\nu^{m+1}(k_1, \ldots, k_m, k) + v_k^2 R_{\nu+1}^{m+1}(k_1, \ldots, k_m, k). \tag{11.70}$$

* A quite different method to treat particle number violation by two constraints on $\hat{N}$ and $\hat{N}^2$ has been proposed by Lipkin and Nogami [Li 60b, No 64b, NZ 64, SR 78].

The projected expectation value of the energy takes the form*

$$E^{A}_{\text{proj}} = \frac{\langle \Psi^A | H | \Psi^A \rangle}{\langle \Psi^A | \Psi^A \rangle}$$

$$= R_0^{0^{-1}} \cdot \Big\{ \sum_{k \gtrless 0} \epsilon_k v_k^2 R_1^1(k) + \frac{1}{2} \sum_{kk' \gtrless 0} \bar{v}_{kk'kk'} v_k^2 v_{k'}^2 R_2^2(k,k') + \sum_{k,k'>0} \bar{v}_{k\bar{k}k'\bar{k}'} u_k v_k u_{k'} v_{k'} R_1^2(k,k') \Big\}. \quad (11.71)$$

We see that the expression (11.71) for the projected energy in this formalism is very similar to the analogous formula for the BCS energy (6.45). In the same way, we can calculate expectation values of arbitrary operators with particle number projected BCS functions (usually this method is called *PBCS theory*).

A bit more complicated is the *FBCS theory*, a variation after particle number projection. In this case, we must vary the projected energy with respect to the parameters v_k^2. Proceeding in the same way as we have in the derivation of the BCS equations (6.48), we obtain a set of so-called FBCS equations [DMP 64]. They have a structure similar to the ordinary BCS equations and their solution provides no further difficulties.

Table 11.1 shows tota! energies of valence nucleons for even–even systems in the actinide region. As expected, the projected energies are always deeper by several hundred keV. The major part already comes from the elimination of spurious components in the BCS wave function by the PBCS procedure. The variation in the FBCS method causes an additional lowering.

The details depend strongly on the level density in the vicinity of the fermi surface. In general, with decreasing pairing correlations (i.e., with decreasing level density, which happens in Table 11.1 for increasing proton numbers) we observe an increase in the error of the PBCS method. In other words, for large correlations with strong symmetry violations, the mean field solution, which is obtained by a symmetry conservation on the average (in our case the simple BCS wave function) already provides a good approximation of the solution of the projected variational

Table 11.1. Total energies of valence nucleons for even proton systems in the actinide region calculated by different BCS techniques (25 orbitals outside a $Z=76$ core). (From [MPR 65].)

Z	$-E_{\text{FBCS}}$ [MeV]	$E_{\text{BCS}}-E_{\text{FBCS}}$ [keV]	$E_{\text{PBCS}}-E_{\text{FBCS}}$ [keV]
88	91.433	264	2
90	106.005	330	28
92	120.340	400	36
94	134.312	427	51
96	147.956	553	72
98	160.829	494	84
100	173.210	496	77

*Residues with negative k-values should be set equal to the ones with the same positive indices in Eq. (11.71). For equal indices, we also have to use the rule

$$R_2^2(kk) = R_1^1(k).$$

problem. For small correlations the mean field approach does not give a satisfying internal wave function.

In particular, we know that below a critical level density (which is equivalent to a critical force strength G_c) the gap equation (6.60) only has the trivial solution with vanishing gap. This is no longer true in the FBCS method. It can also account for rather weak pairing correlations. As a measure of these correlations, we can no longer use the parameter Δ in a projected theory, since it describes a "deformation," which vanishes in the symmetry conserving wave function. Instead, we can use, for instance, the function

$$D^2 = \sum_{k,k'>0} \langle a_k^+ a_{\bar{k}}^+ a_{\bar{k}'} a_{k'} \rangle, \tag{11.72}$$

which measures the number of Cooper pairs in the wave function. D coincides to leading order with $\Delta/\sqrt{G}$ in the BCS theory. In the projected theory it is given by

$$D^2 = \sum_{kk'>0} R_1^2(k,k') \frac{u_k v_k u_{k'} v_{k'}}{R_0^0}. \tag{11.73}$$

Figure 11.5 shows the qualitative behavior of this quantity as a function of the coupling strength G. In the BCS and PBCS cases, the pairing correlation vanishes for $G < G_c$. The BCS solution, in particular, shows a rather sharp cusp at $G = G_c$. This sharp "phase transition," however, is washed out for finite systems. It disappears in the FBCS model and we find a gradual weakening of the pairing correlations as a function of G. In all cases of weak pairing correlations we should therefore go beyond PBCS theory. An example is calculations in the high spin region, where we expect a drastic reduction of the pairing field by either a decoupling mechanism or the Mottelson–Valatin effect in the rotating system (see Sec. 7.7).

As we saw in Sec. 7.7, in these cases we need the full HFB theory, that is, we also have to vary with respect to the single-particle wave functions φ_k. *Particle number projected HFB theory* after the variation can be obtained rather simply from the method discussed so far. Since general HFB wave functions in the canonical basis (see Sec. 7.2) have the structure of BCS functions, we have only to transform to the canonical basis and obtain in the same way the projected matrix elements.

An approximate method that avoids these transformations is based on the idea of Fomenko to replace the integral in Eq. (11.66) by a sum over L points $\zeta_l = \exp(2\pi i l/L)$ on the unity circle [Fo 70, So 72]. The approximate projector then

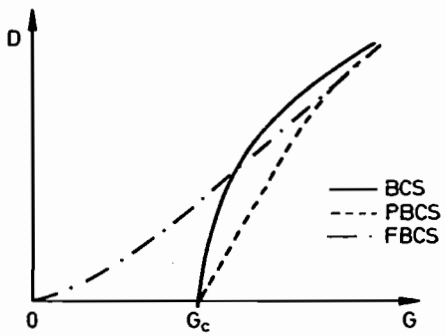


Figure 11.5. Pairing correlations as a function of the coupling strength G in the BCS (full), PBCS (dashed), and FBCS (dashed/dotted) method (qualitatively).

has the form

$$\hat{P}_L^A = \frac{1}{L} \sum_{l=1}^{L} (\zeta_l)^{(\hat{N}-A)/2}. \tag{11.74}$$

It is easy to see that $\hat{P}^{A+2m} \cdot \hat{P}_L^A |\mathrm{BCS}\rangle$ vanishes unless $m=0, \pm L, \pm 2L, \ldots$. This means that the wave function $\hat{P}_L^A |\mathrm{BCS}\rangle$ contains only components with the particle numbers A, $A \pm 2L$, $A \pm 4L, \ldots$. For instance, the approximate projector $\hat{P}_2^A$ already annihilates the components with $A \pm 2, A \pm 6, A \pm 10, \ldots$. The projected wave function ($A=2p$)

$$\hat{P}_L^{2p} |\mathrm{BCS}\rangle = \frac{1}{L} \sum_{l=1}^{L} \zeta_l^{-p} \prod_{k>0} (u_k + v_k \zeta_l a_k^+ a_{\bar{k}}^+) |-\rangle \tag{11.75}$$

is a linear combination of L BCS wave functions rotated in the gauge space by the angle $\varphi_l = 2\pi l / L$; that is, v_k is replaced by $e^{i\varphi_l} \cdot v_k$. In the same way $\hat{P}_L^A$ transforms an HFB function $|\Phi\rangle$ given by the coefficients $(U_{kk'}, V_{kk'})$ into a linear combination of HFB functions $|\Phi_l\rangle$ $(l=1, \ldots, L)$ with the coefficients $(U_{kk'}, \zeta_l V_{kk'})$. A transformation to the canonical basis is not necessary.

Since the BCS wave function is smeared out only over a few particle numbers, for small values of L, we already obtain a nearly exact projection.

A full HFB theory with variation after particle number projection has so far not been carried out. In any case, it is much more complicated than FBCS. In all practical calculations functions $|\Phi(\beta, \gamma, \Delta)\rangle$ depending on a limited number of parameters have been used to minimize the projected energy.

11.4.4 Approximate Projection for Large Deformations

A quite general method for approximate projection for the case of strong symmetry violations has been given by Kamlah [Ka 68] for the case of the rotational group (see also [Kü 72]). Since the rotational group shows many complicated special features, and since the method is also applicable to other cases, such as the particle number [DMR 75, Ma 75a], we shall present it here for the case of a one-dimensional rotation around a fixed axis only. It will be described by the operator $\hat{J}$ and the angle α.

The projector onto eigenfunctions of $\hat{J}$ has the form (11.57)

$$\hat{P}^I = \frac{1}{2\pi} \int_0^{2\pi} e^{i\alpha(\hat{J}-I)} \, d\alpha \tag{11.76}$$

and for the projected energy we obtain

$$E_{\mathrm{proj}}^I = \frac{\langle \Phi | H \hat{P}^I | \Phi \rangle}{\langle \Phi | \hat{P}^I | \Phi \rangle} = \frac{\int d\alpha \, h(\alpha) e^{-iI\alpha}}{\int d\alpha \, n(\alpha) e^{-iI\alpha}} \tag{11.77}$$

with overlap integrals

$$\begin{Bmatrix} h(\alpha) \\ n(\alpha) \end{Bmatrix} = \langle \Phi | \begin{Bmatrix} H \\ 1 \end{Bmatrix} e^{i\alpha \hat{J}} | \Phi \rangle. \tag{11.78}$$

In the case of large particle numbers and strong deformations, we expect the functions $n(\alpha)$ and $h(\alpha)$ to be sharply peaked at $\alpha=0$ and to vanish elsewhere in such a way that the quotient $h(\alpha)/n(\alpha)$ is a rather smooth function. This is a general property of many-body wave functions (which was discussed in great detail in Sec. 10.7). Following the arguments of Section 10.7.4, we obtain*

$$n(\alpha)\simeq\exp\left(i\langle\hat{J}\rangle\cdot\alpha-\tfrac{1}{2}\langle\Delta\hat{J}^2\rangle\alpha^2\right), \tag{11.79}$$

with

$$\Delta\hat{J}=\hat{J}-\langle\hat{J}\rangle,$$

the only difference now being that we also allow for time odd components in the wave functions, which does not give a pure Gaussian, but an additional phase in Eq. (11.79) ($\langle\hat{J}\rangle\neq 0$). The idea of Kamlah is now to use the operator

$$\hat{I} := -\langle\hat{J}\rangle+\frac{1}{i}\frac{\partial}{\partial\alpha},$$

which is a representation of the symmetry operator in the space of the parameter α,† and to expand the Hamilton overlap function $h(\alpha)$ in the following way.

$$h(\alpha)=\sum_{n=0}^{M} h_n\hat{I}^n n(\alpha). \tag{11.80}$$

Mathematically this essentially corresponds to a Taylor expansion of the Fourier transformed function $h(\alpha)/n(\alpha)$. Since we need $h(\alpha)$ only in the vicinity of $\alpha=0$ we can already hope to get a very good approximation for this function with a few non-vanishing constants $h_0, h_1,\ldots,h_M$, if they are properly adjusted. The number M certainly depends on the width of the overlap (11.79), that is, on the fluctuation $\langle\Delta\hat{J}^2\rangle$ of the symmetry operator $\hat{J}$ in the wave function $|\Phi\rangle$. The larger the symmetry violation, the better the approximation will already be for low M-values.

The expansion (11.80) is not a perturbative expansion in the sense that the coefficients h_n are determined step by step one after the other, such that the coefficient h_0 does not change going from M to $M+1$, but for each M-value the whole set $h_0,\ldots,h_M$ is calculated at once: Operating with $1, \hat{I}, \hat{I}^2,\ldots,\hat{I}^M$ on Eq. (11.80) we obtain, in the limit $\alpha\to 0$, the following inhomogeneous system for the unknowns $h_0,\ldots,h_M$:

$$\langle H(\Delta\hat{J})^m\rangle=\sum_{n=0}^{M} h_n\langle(\Delta\hat{J})^{m+n}\rangle. \tag{11.81}$$

The solution of these equations gives us an expression for the projected

* In the following we use the symbol $|\,\rangle$ for the normalized intrinsic function $|\Phi\rangle$.

† In the case of three-dimensional rotations we therefore use, instead of $(1/i)\partial/\partial\alpha$, the representation of the angular momenta in Euler angles (see Appendix A).

energy. From (11.77) and (11.80), we get

$$E^I_{\text{proj}} = \sum_{n=0}^{M} h_n \cdot (I - \langle \hat{J} \rangle)^n. \tag{11.82}$$

The simplest case is $M = 1$. Here we find

$$h_0 = \langle H \rangle, \qquad h_1 = \frac{\langle H \Delta \hat{J} \rangle}{\langle \Delta \hat{J}^2 \rangle}, \tag{11.83}$$

and

$$E^I_{\text{proj}} = \langle \Phi | H - h_1 \hat{J} | \Phi \rangle + h_1 \cdot I. \tag{11.84}$$

In a variation after projection method we get the variational equation

$$\frac{\delta}{\delta \Phi} \langle \Phi | H | \Phi \rangle - h_1 \frac{\delta}{\delta \Phi} \langle \Phi | \hat{J} | \Phi \rangle + (I - \langle \hat{J} \rangle) \frac{\delta h_1}{\delta \Phi} = 0. \tag{11.85}$$

It is solved by a solution of the cranking equations*

$$\frac{\delta}{\delta \Phi} \langle \Phi | H | \Phi \rangle - \omega \frac{\delta}{\delta \Phi} \langle \Phi | \hat{J} | \Phi \rangle = 0, \tag{11.86}$$

with

$$\langle \hat{J} \rangle = I$$

if the condition $h_1 = \omega$ is fulfilled. To see that this is always the case we use the fact that Eq. (11.86) has to hold for arbitrary variations $|\delta \Phi\rangle$ in the set of product wave functions. One such variation is given by $|\delta \Phi\rangle \propto \Delta \hat{J} |\Phi\rangle = \hat{J}^{20} |\Phi\rangle$ (in the sense of a quasi-particle representation of Chap. 7). Therefore, we get

$$\langle \Delta \hat{J} H \rangle - \omega \langle (\Delta \hat{J})^2 \rangle = 0,$$

which shows that $\omega = h_1$ in Eq. (11.83).

We now have proven the remarkable result that in the case of strong symmetry violations, where $M = 1$ is a sufficient approximation in the expansion (11.80), cranking model wave functions that preserve the average symmetry are a solution of the variational equations after projection. This result does not depend on the structure of the overlap functions in detail. One only needs the fact that they are sharply peaked around the origin.

So far we have shown this only in the case of a one-dimensional rotation, but the result (as we shall see in Sec. 11.4.4) can be transferred with minor changes also to the three-dimensional case.

Within this approximation, the projected energy is given by h_0 and equal to the unprojected energy. This justifies the self-consistent cranking model calculations for the rotational spectra (see Sec. 7.7). In particular, for the

* In some cases it turns out that this solution does not correspond to a minimum but to a flat maximum. Second-order terms ($M = 2$), however, guarantee that this maximum is changed to a minimum [BMR 70].

self-consistent moment of inertia we find

$$\mathcal{I}_{sc}^{-1} = \frac{\omega}{\langle \hat{J} \rangle} = \frac{\langle H \Delta \hat{J} \rangle}{\langle \Delta \hat{J}^2 \rangle \langle \hat{J} \rangle}. \tag{11.87}$$

As discussed in Section 8.5.4, it is at the point $\omega = 0$ identical with that of Thouless and Valatin.

This simple result, however, only applies for the energy expectation value. If we want to calculate moments and transition probabilities, we have to carry out an explicit projection from the intrinsic cranking functions (see Sec. 11.4.6.4).

In cases where the restriction to $M = 1$ in the expansion (11.80) is not good enough, we have to go to higher M-values. For $M = 2$ we neglect* $\langle \Delta \hat{J}^3 \rangle$, and get

$$h_0 = \langle H \rangle - \frac{\langle \Delta \hat{J}^2 \rangle}{2\mathcal{I}_Y}, \qquad h_1 = \frac{\langle H \Delta \hat{J} \rangle}{\langle \Delta \hat{J}^2 \rangle} = \frac{\langle \hat{J} \rangle}{\mathcal{I}_{sc}}, \qquad h_2 = \frac{1}{2\mathcal{I}_Y}, \tag{11.88}$$

with

$$\frac{1}{2\mathcal{I}_Y} = \frac{\langle H \Delta \hat{J}^2 \rangle - \langle H \rangle \langle \Delta \hat{J}^2 \rangle}{\langle \Delta \hat{J}^4 \rangle - \langle \Delta \hat{J}^2 \rangle^2}. \tag{11.89}$$

In the case of a Gaussian norm (11.79), we have $\langle \Delta \hat{J}^4 \rangle = 3 \langle \Delta \hat{J}^2 \rangle^2$ and find

$$\mathcal{I}_Y^{-1} = \frac{\langle (H - \langle H \rangle) \Delta \hat{J}^2 \rangle}{\langle \Delta \hat{J}^2 \rangle^2}. \tag{11.90}$$

The projected energy now has the form:

$$E^I_{\text{proj}} = \langle H \rangle - \frac{\langle \Delta \hat{J}^2 \rangle}{2\mathcal{I}_Y} + \frac{\langle \hat{J} \rangle}{\mathcal{I}_{sc}} (I - \langle \hat{J} \rangle) + \frac{1}{2\mathcal{I}_Y} (I - \langle \hat{J} \rangle)^2. \tag{11.91}$$

Let us first study the method of *variation before projection*, which was originally proposed by Peierls and Yoccoz [PY 57, Yo 57]. Here the wave function $|\Phi\rangle$ is obtained from a minimization of H without constraint. It is, therefore, time-reversal invariant and has vanishing expectation values $\langle \hat{J} \rangle$ and $\langle H \hat{J} \rangle$. The spectrum then has the form

$$E^I_{\text{proj}} = \langle H - \frac{\hat{J}^2}{2\mathcal{I}_Y} \rangle + \frac{I^2}{2\mathcal{I}_Y}. \tag{11.92}$$

It is the spectrum of a one-dimensional rotor with the Yoccoz value $\mathcal{I}_Y$ for the moment of inertia (see Sec. 11.4.5).† The band head is obtained from the HF energy $\langle H \rangle$ by subtracting the spurious rotational energy in the

* $\langle \Delta \hat{J}^3 \rangle$ measures the asymmetry of the I distribution in $|\Phi\rangle$. In the case of a Gaussian norm overlap, it vanishes exactly.

† The method has also been investigated in great detail by Verhaar [Ve 63, Ve 64] for the three-dimensional rotation in even and odd nuclei for $K = 0$ and $K \neq 0$ bands. Thus one can derive microscopic expressions for the B and C coefficients in Eq. (3.1) as well as for the decoupling parameters (3.31).

symmetry-violating wave function.* This produces a substantial lowering of the ground state energy. In well deformed nuclei this term is of the order of 4–6 MeV. We have already encountered this term several times, for instance, in the method of redundant coordinates (see Sec. 11.3.2) and in the treatment of symmetry violations within the RPA approach (Sec. 8.4.8).

Carrying out a *variation after projection*, we have to minimize the r.h.s. of Eq. (11.91) with respect to $|\Phi\rangle$. In the case where the spurious energy $\langle\Delta\hat{J}^2\rangle/2\vartheta_Y$ does not depend very much on $|\Phi\rangle$ we again find that the cranking model is a solution. Model calculations [BMR 70] in well-deformed nuclei of the rare earth region have shown that this term indeed gives a rather flat contribution to the energy surface.

The convergence of the Kamlah expansion (11.80) depends on how sharply the overlap functions are peaked at the origin. It is measured by the width $\langle\Delta\hat{J}^2\rangle$, that is, by the fluctuation of the symmetry operator. If this fluctuation is large, the expansion converges rapidly. We call such cases the limit of *large deformations*, and we want to emphasize that this limit is not necessarily connected with large deformation parameters of the corresponding mean field. In the case of large particle numbers, the fluctuation $\langle\Delta\hat{J}^2\rangle$ can have a very large value, even for small deformation parameters. In fact, the shape of "well deformed" heavy nuclei in the ground state does not differ very much from a sphere (see, for instance, Fig. 7.5). Because of the large particle numbers, however, the overlap functions are strongly peaked at the origin, which indicates that the assumption of a phase transition to a deformed mean field is well justified.

11.4.5 The Inertial Parameters

In all collective motions connected with the violation of a continuous symmetry, the potential energy $\langle H\rangle$ (11.9) does not depend on the collective coordinate. Therefore, there are no restoring forces and in most cases of the rotational type we always get motions with large amplitude. The crucial parameters of the collective Hamiltonian are therefore the inertial parameters.

There is one case where we know this parameter a priori: Because of Galilean invariance, the collective inertia of *translations* have to be equal to the total mass Am of the system. This case therefore provides a very good test of the different approximation schemes used so far in the treatment of symmetry violations.

We start with the method of *variation after projection* [Ze 65, Vi 66, RY 66]. Equation (11.62) defines the intrinsic solution for a wave function with momentum $\mathbf{p}$:

$$\langle\delta\Phi|(H-E^{\mathrm{p}})\hat{P}^{\mathrm{p}}|\Phi\rangle=0. \tag{11.93}$$

* For calculations concerning this contribution, see [ZN 77, VLS 79, ER 80].

Transforming from the laboratory system to a system which moves with the velocity $\mathbf{v}=\mathbf{p}/2Am$, we find ($\mathbf{R}$ is the center of mass coordinate):

$$|\Phi\rangle = e^{(i/\hbar)\mathbf{pR}}|\Phi'\rangle \tag{11.94}$$

and

$$\langle\delta\Phi'|e^{-(i/\hbar)\mathbf{pR}}(H-E^{\mathbf{p}})\hat{P}^{\mathbf{p}}e^{(i/\hbar)\mathbf{pR}}|\Phi'\rangle = 0. \tag{11.95}$$

Using the definition (11.59) of the projector $\hat{P}^{\mathbf{p}}$, we obtain the transformation

$$e^{-(i/\hbar)\mathbf{pR}}\hat{P}^{\mathbf{p}}e^{(i/\hbar)\mathbf{pR}} = \hat{P}^{\mathbf{0}}. \tag{11.96}$$

For a local two-body interaction we get

$$e^{-(i/\hbar)\mathbf{pR}}He^{(i/\hbar)\mathbf{pR}} = H + \frac{\mathbf{pP}}{Am} + \frac{\mathbf{p}^2}{2Am}. \tag{11.97}$$

Since $\hat{P}^{\mathbf{0}}$ projects onto states with vanishing eigenvalue of the center of mass momentum $\mathbf{P}$, Eq. (11.95) has the form

$$\langle\delta\Phi'|\left(H-E^{\mathbf{p}}+\frac{\mathbf{p}^2}{2Am}\right)\hat{P}^{\mathbf{0}}|\Phi'\rangle = 0. \tag{11.98}$$

It corresponds to the variational problem for vanishing momentum and energy $E^{\mathbf{0}}$. The energy of the system with momentum $\mathbf{p}$ is therefore given by

$$E^{\mathbf{p}} = E^{\mathbf{0}} + \frac{\mathbf{p}^2}{2Am}, \tag{11.99}$$

which shows that we obtain the proper total mass using the method of projection after variation. This property is conserved if we approximate the projection after variation by the *self-consistent cranking model* [Gr 60] (in the case of translations, we usually say "pushing" model):

$$\langle\delta\Phi|H-\mathbf{vP}|\Phi\rangle = 0. \tag{11.100}$$

The inertia parameter of Thouless and Valatin (8.113) obtained in this way is the total mass. In principle, we have shown this already in Section 8.4.7, but it is easy to verify it once more if we use the admissible variation

$$|\delta\Phi\rangle \propto \mathbf{R}|\Phi\rangle. \tag{11.101}$$

Subtracting Eq. (11.100) from its complex conjugate, we obtain

$$\langle\Phi|[H,\mathbf{R}]-\mathbf{v}\cdot[\mathbf{P},\mathbf{R}]|\Phi\rangle = 0. \tag{11.102}$$

In the case of H containing only a local two-body interaction, we get

$$\mathbf{v} = \frac{\langle\Phi|\mathbf{P}|\Phi\rangle}{Am}, \tag{11.103}$$

which shows that the inertia parameter, which is given by the ratio of the expectation value of $\mathbf{P}$ to the velocity $\mathbf{v}$, is again the total mass.

We can deduce the simple *Inglis formula* [see Eq. (3.89)] as the inertia parameter also from the variational equation (11.93) if we replace the

two-body interaction in the Hamiltonian H by a one-body potential (the variational principle in this case is equivalent to an exact diagonalization). This shows us that for a local one-body potential as, for instance, for a Woods–Saxon well, the Inglis formula also gives the exact total mass* [Vi 63].

It is a drawback of the method of *variation before projection*—which yields the inertia of Yoccoz† (11.90)—that it violates the Galilean invariance. It does not produce the exact mass. From Eqs. (11.90) and (10.134) we obtain for the inertia in the notation of Section 8.4.7 (see also [FW 70]):

$$\mathcal{I}_Y^{-1} = \frac{1}{4\langle \Delta P^2\rangle^2}\langle [P,[H,P]_+]_+\rangle_L = \frac{(P^*P)\begin{pmatrix} A & B \\ B^* & A^* \end{pmatrix}\begin{pmatrix} P \\ P^* \end{pmatrix}}{\left(2\sum_{mi}|P_{mi}|^2\right)^2}, \tag{11.104}$$

where P is the *ph* part of one component of the vector $\mathbf{P}$ if the self-consistent basis ($H^{20}=0$) is used. In the limit of a vanishing residual interaction, it is equal to

$$\mathcal{I}_Y(V=0) = \frac{2\left(\sum_{mi}|P_{mi}|^2\right)^2}{\sum_{mi}|P_{mi}|^2(\epsilon_m-\epsilon_i)}. \tag{11.105}$$

It is obviously different from the Thouless–Valatin inertia (8.108),

$$\mathcal{I}_{TV}^{-1} = \frac{1}{\hbar^2}\langle [R,[H,R]_-]_-\rangle = \frac{2}{\hbar^2}S_1(R)$$

or

$$\mathcal{I}_{TV} = \frac{1}{2}(P^*P)\begin{pmatrix} A & B \\ B^* & A^* \end{pmatrix}^{-1}\begin{pmatrix} P \\ P^* \end{pmatrix} \tag{11.106}$$

which in the limit of vanishing interaction goes over into the Inglis formula (3.89),

$$\mathcal{I}_{TV}^{-1}(V=0) = 2\sum_{mi}\frac{|P_{mi}|^2}{\epsilon_m-\epsilon_i}. \tag{11.107}$$

*This is not a contradiction to the fact that the Thouless–Valatin mass (8.113) differs from the Inglis formula by the residual interaction, because for all forces that produce a local HF field, the residual interaction has no influence on the mass (the time odd part Γ_1 in Eq. (12.81) vanishes).

†The idea that the intrinsic wave function $|\Phi\rangle$ is a superposition of all the members of the rotational band Ψ^I has also been used by Skyrme [Sk 57]: Assuming that the intrinsic Hamiltonian $H'=H-\mathbf{J}^2/2\mathcal{I}$ has an eigenstate, which is not an eigenstate of $\mathbf{J}^2$, we can show [Li 60a] that H has a rotational spectrum with $\mathcal{I}$ as moment of inertia. To determine the optimal value of $\mathcal{I}$, Skyrme minimized the expectation value $\langle (H'-\langle H'\rangle)^2\rangle$ with respect to the parameter $\mathcal{I}$, and found as a solution

$$\frac{1}{2\mathcal{I}_{\text{Skyrme}}} = \frac{\langle (H-\langle H\rangle)\mathbf{J}^2\rangle}{\langle \mathbf{J}^4\rangle - \langle \mathbf{J}^2\rangle^2}.$$

This expression is for time-even wave functions identical with Eq. (11.89), and therefore closely related to $\mathcal{I}_Y$ (see also [Wo 75]).

How large and how important this difference actually is, has not been investigated very much. It certainly depends on the interaction and on the single-particle energies ϵ_k. For all models with a constant level spacing $\epsilon = \epsilon_m - \epsilon_i$, both values are identical. In Sec. 10.7 we saw that we have the Yoccoz inertia (11.104) for all kinds of collective motions, if we use the simple GCM method with coordinate dependent generating functions. Only in theories in which we allow the intrinsic function $|\Phi\rangle$ to also have time-odd components that depend on the momentum do we eventually obtain the proper mass. This is the case in the *time-dependent Hartree–Fock method* (see Chap. 12) and in the variation-after-projection method.

To obtain the proper mass in a GCM method also, *Thouless and Peierls* [PT 62] proposed to superpose not only wave functions of different positions [see Eq. (11.59)],

$$|\Phi(\mathbf{q})\rangle = e^{(i/\hbar)\mathbf{q}\mathbf{P}}|\Phi\rangle, \tag{11.108}$$

but also those with different velocities,

$$|\Phi(\mathbf{q},\mathbf{v})\rangle = e^{-(i/\hbar)M\mathbf{v}\mathbf{R}}|\Phi(\mathbf{q})\rangle. \tag{11.109}$$

This yields a double projection method, in which the velocity dependence of the weight function $f(\mathbf{q},\mathbf{v})$ has to be determined by variation. We can show that the wave functions obtained in this way do not violate Galilean invariance. The method itself, however, is rather complicated and up to now has not been applied very much [Ba 69b]. In practical calculations the method of variation after projection is usually preferred (see Sec. 11.4.2).

So far we have treated only the case of translations. In the case of rotations or other collective motions as discussed in Section 10.7 the situation is more complicated. The exact inertia is not known a priori. We can only argue in analogy to the translational case. Numerical calculations have shown, at least, that in the rotational case, the moments of inertia obtained by the method of Thouless and Valatin are, with proper handling of the pairing correlations, in good agreement with experimental data (see Fig. 3.14).

11.4.6 Angular Momentum Projection

11.4.6.1. Derivation of Projector-like Operators. The rotational group is usually parametrized by the three Euler angles* $\Omega = (\alpha, \beta, \gamma)$, and the general rotation is given in Eq. (A.4). We have to go back to the ansatz (11.53) of the wave function $|\Psi\rangle$. As in Eq. (11.55), we expand the weight function $f(\Omega)$ into a complete set of eigenfunctions of the corresponding symmetry operators: the angular momenta represented in Euler angles. As

*There have been used other representations also, for instance

$$R = \exp(i\varphi\hat{\mathbf{J}})$$

for a general rotation with the parameters $\varphi_x, \varphi_y, \varphi_z$ [KO 77].

discussed in Appendix A, such a set is given by the Wigner functions $D^{I*}_{MK}(\Omega)$. We therefore gain the ansatz

$$f(\Omega)=\frac{2I+1}{8\pi^2}\sum_{IMK} g_{IMK} D^{I*}_{MK}(\Omega) \tag{11.110}$$

and

$$|\Psi\rangle=\sum_{IMK} g_{IMK}\hat{P}^I_{MK}|\Phi\rangle. \tag{11.111}$$

The operators [RW 54, PY 57, Vi 66]

$$\hat{P}^I_{MK}=\frac{2I+1}{8\pi^2}\int D^{I*}_{MK}(\Omega)R(\Omega)\,d\Omega \tag{11.112}$$

$$R(\Omega)=e^{i\alpha\hat{J}_z}e^{i\beta\hat{J}_y}e^{i\gamma\hat{J}_z}$$

can be written in analogy to (11.58) [LB 68, Co 71]

$$\hat{P}^I_{MK}=\sum_{\alpha}|IM\alpha\rangle\langle IK\alpha|, \tag{11.113}$$

where $|IM\alpha\rangle$ is a complete orthonormalized set in the Hilbert space and α characterizes all quantum numbers besides the rotational quantum numbers I and M.

From (11.113) we see that $\hat{P}^I_{MK}$ are not projectors in the mathematical sense. We only have the relations

$$\hat{P}^I_{MK}\hat{P}^{I'}_{M'K'}=\delta_{II'}\delta_{M'K}\hat{P}^I_{MK'};\qquad (\hat{P}^I_{MK})^+=\hat{P}^I_{KM}. \tag{11.114}$$

To show how they act on the "internal"* symmetry-violating wave function $|\Phi\rangle$, we decompose $|\Phi\rangle$ with respect to eigenstates of the operator $\hat{J}_z$ with quantum number K:

$$|\Phi\rangle=\sum_{K'}|\Phi_{K'}\rangle;\qquad \hat{J}_z|\Phi_K\rangle=K|\Phi_K\rangle. \tag{11.115}$$

$\hat{P}^I_{MK}$ transforms the component $|\Phi_K\rangle$ into an eigenfunction of $\hat{\mathbf{J}}^2$ and $\hat{J}_z$ with the quantum numbers I and M and annihilates all other components with $K'\neq K$.

In principle it is very simple to construct the precise mathematical projector onto the quantum numbers I, M. It is given by†

$$\hat{P}^I_{MM}=\sum_{\alpha}|IM\alpha\rangle\langle IM\alpha|. \tag{11.116}$$

*Actually, in the GCM method we introduce neither an intrinsic frame nor redundant coordinates. The connection between the "internal" wave function $|\Phi\rangle$ and the function $|\Psi\rangle$ in the laboratory frame is given not by a coordinate transformation, but by the GCM ansatz (11.111).

†This operator has been represented by Löwdin [Lö 64] in the following way.

$$\hat{P}^I_{MM}=\prod_{L\neq I}\frac{\hat{\mathbf{J}}^2-L(L+1)}{I(I+1)-L(L+1)}\prod_{K\neq M}\frac{\hat{J}_z-K}{M-K}.$$

In practical applications of this operator, we end up with a set of algebraic equations (see also [KL 64, Ke 66, Ul 71, 72, MKL 71, MM 76]) and do not have to perform integrals over the Euler angles.

From (11.115) we see, however, that this operator is rather unphysical. Acting, for instance, on an axial symmetric Nilsson function with $K=0$, we would get only wave functions $|\Psi^{IM}\rangle$ with $M=0$, which clearly violates the rotational symmetry.

We therefore have to use the general ansatz*

$$|\Psi^{IM}\rangle = \sum_K g_K \hat{P}^I_{MK}|\Phi\rangle = \sum_K g_K \frac{2I+1}{8\pi^2}\int d\Omega\, D^{I*}_{MK}(\Omega) R(\Omega)|\Phi\rangle. \quad (11.117)$$

In cases where the wave function $|\Phi\rangle$ has axial symmetry, only the K-value K_0 contributes to the sum (11.117), and the coefficient g_{K_0} is given by the normalization. In general, however, we have to minimize the Hamiltonian with respect to the coefficients g_K. The energy is given by

$$E^I_{\text{proj}} = \frac{\langle\Psi^{IM}|H|\Psi^{IM}\rangle}{\langle\Psi^{IM}|\Psi^{IM}\rangle} = \frac{\sum_{KK'} g^*_K g_{K'} h^I_{KK'}}{\sum_{KK'} g^*_K g_{K'} n^I_{KK'}}, \quad (11.118)$$

with the kernels

$$\begin{aligned} h^I_{KK'} &= \langle\Phi|H\hat{P}^I_{KK'}|\Phi\rangle, \\ n^I_{KK'} &= \langle\Phi|\hat{P}^I_{KK'}|\Phi\rangle. \end{aligned} \quad (11.119)$$

The coefficients g_K are the solution of the generalized eigenvalue problem

$$\sum_{K'} h^I_{KK'} g_{K'} = E^I \sum_{K'} n^I_{KK'} g_{K'}. \quad (11.120)$$

It corresponds to a diagonalization of the Hamiltonian H in the space spanned by the nonorthogonal system $\hat{P}^I_{MK}|\Phi\rangle$.

In Section 10.7 we saw how to proceed in order to end up with a Hermitian eigenvalue problem. We must either orthogonalize the set $\hat{P}^I_{MK}|\Phi\rangle$ or calculate the square root of the matrix $n^I_{KK'}$. In the case of a Gaussian form of the overlap function $\langle\Phi|R(\Omega)|\Phi\rangle$, that is, for a strongly deformed wave function (see Sec. 11.4.4), we can do this analytically. Since we are faced with a three dimensional problem, we end up with a collective Hamiltonian like the one of Eq. (10.145). Une et al. [UIO 76] have shown that it has exactly the form of the rotor Hamiltonian (1.55) with moments of inertia given by the Yoccoz formula† (see Sec. 11.4.5).

The close analogy to the collective rotor model can be visualized rather easily if we define extremely deformed many-body wave functions‡ $|\Omega\rangle$, which point like a needle in the direction Ω. They are nearly localized in Ω-space. The overlap between these functions and the wave function $|\Psi^{IM}\rangle$ (11.117),

$$\langle\Omega|\Psi^{IM}\rangle \propto \sum_K g_K \int d\Omega'\, D^{I*}_{MK}(\Omega')\langle\Omega|R(\Omega')|\Phi\rangle, \quad (11.121)$$

* The coefficients g_K will certainly also depend on I, as we see from Eq. (11.120). However, we omit the index I.

† See also [ABC 77a].

‡ These functions correspond to the biorthogonal basis discussed in Eq. (10.20)ff.

measures the probability amplitude of finding the system at the angles Ω. In the limit where the wave function $|\Phi\rangle$ is well deformed, peaking at $\Omega=0$, the overlap $\langle\Omega|R(\Omega')|\Phi\rangle$ behaves like the δ-function $\delta(\Omega-\Omega')$ in Ω-space, and we get [Ze 67, FJM 71]

$$\langle\Omega|\Psi^{IM}\rangle \propto \sum_K g_K D^{I*}_{MK}(\Omega), \tag{11.122}$$

which is exactly the wave function of a rotor (see Sec. 1.5.1). In this approximation, the deformed function $|\Phi\rangle$ moves in space like a rotor. We have to emphasize, however, that we use no redundant variables. The Euler angles Ω are only parameters used as generator coordinates.

In the same way as the symmetry behavior of the rotor, under certain operations of the group D_2 [as, for instance, $\mathcal{R}_1$ or $\mathcal{R}_2$ in Eqs. (1.57)ff.] causes relationships between the expansion coefficients $g_K^{(\text{rotor})}$, we can now derive corresponding relations for the coefficients g_K if our internal wave function has these symmetries.

As an example, we use a rotation of 180° around the x-axis. We assume* that the internal wave function $|\Phi\rangle$ is an eigenfunction with respect to this operation:

$$e^{i\pi\hat{J}_x}|\Phi\rangle = r_x|\Phi\rangle. \tag{11.124}$$

The eigenvalue r_x is ± 1 for systems with even and $\pm i$ for systems with odd particle number. It is often called the *signature* [Bo 76b]. The ground state of an even–even nucleus in a deformed Nilsson well, for instance, always has $r_x=+1$, because a pair-wise occupation of the degenerate levels with $\pm i$ gives the lowest energy. The condition $r_x=+1$ for the ground state band is therefore a natural consequence of the variational principle. This fact provides a microscopic foundation for the assumption (1.58) of the rotor model—that the deformed core has the eigenvalue $+1$ with respect to the operation $\mathcal{R}_1$ (1.57).

The symmetry (11.123) is rather general and also applies for cranking model wave functions. So far, no experimental evidence has been found that it is not valid. We shall therefore use this symmetry in the following.

If we decompose the internal wave function $|\Phi\rangle$ into its components $|\Phi_K\rangle$ with good K-quantum number, as in Eq. (11.115), we get from (11.123):

$$e^{i\pi\hat{J}_x}|\Phi_K\rangle = r_x|\Phi_{-K}\rangle. \tag{11.125}$$

From Eq. (11.124) we can deduce

$$\hat{P}^I_{MK}e^{i\pi\hat{J}_x} = (-)^I\hat{P}^I_{MK}, \tag{11.126}$$

which shows that the states

$$\hat{P}^I_{MK}|\Phi\rangle = \hat{P}^I_{MK}|\Phi_K\rangle \quad \text{and} \quad \hat{P}^I_{M-K}|\Phi\rangle = \hat{P}^I_{M-K}|\Phi_{-K}\rangle$$

correspond to the same state in the Hilbert space. We can therefore restrict the sum in Eq. (11.117) over $K \geqslant 0$.

*This assumption is compatible with rotational invariance of our method, because a rotation of the function $|\Phi\rangle$ by the Euler angles Ω can be expressed as a transformation of the coefficients g_K, as we may see from the relationship

$$\hat{P}^I_{MK}R(\Omega) = \sum_{K'} D^I_{KK'}(\Omega)\hat{P}^I_{MK'}. \tag{11.123}$$

As long as we take into account arbitrary coefficients g_K, we are free in the choice of the orientation of $|\Phi\rangle$.

The representation

$$|\Psi^{IM}\rangle = \sum_{K>0} \frac{g_K}{1+\delta_{K0}} \left(\hat{P}^I_{MK} + (-)^I r_x^{-1} \hat{P}^I_{M-K}\right)|\Phi\rangle \tag{11.127}$$

clearly shows that an axially symmetric wave function $|\Phi\rangle$ with $K=0$ and $r_x=1$ is projected onto zero for odd I-values. For the ground state band, we therefore get the spin sequence $I=0, 2, 4, \ldots$.

In cases in which we have only approximate axial symmetry, as in the cranking model wave functions (see Secs. 7.7 and 11.4.6.3), the remaining $K\neq 0$ parts are not projected onto zero. They have their origin, however, in admixtures of two-quasiparticle states and more complicated configurations that lie at higher energies and contribute only to highly excited levels in the eigenvalue problem (11.120), in which we are not interested. For the ground state band, we should therefore impose the condition

$$r_x = (-)^I. \tag{11.128}$$

11.4.6.2. The Double Variational Method. So far, we have not specified the method of determining the intrinsic wave function $|\Phi\rangle$. As in the case of Abelian groups discussed in Section 11.4.2, we can carry out a variation either before or after projection. In the case without axial symmetry, the situation is now more complicated, however, since we must also determine the coefficients g_K. In the VBP method, they must be calculated by Eq. (11.120).

In the VAP method, we now end up with a coupled system similar to the system (10.147) [Ze 67]:

$$\begin{aligned} \sum_{K'} g_{K'} \langle\Phi|(H-E^I)\hat{P}^I_{KK'}|\Phi\rangle &= 0, \\ \sum_{KK'} g^*_K g_{K'} \langle\delta\Phi|(H-E^I)\hat{P}^I_{KK'}|\Phi\rangle &= 0. \end{aligned} \tag{11.129}$$

The first part corresponds to Eq. (11.120); the second part is a generalization of the VAP equation (11.62). In principle, this system has to be solved by iteration.

Besides the fact that the system (11.129) is much too complicated to be solved exactly, in many cases it also contains a high degree of redundancy. To see this, we again use the decomposition (11.115) of $|\Phi\rangle$ into its K-components. Using the fact that $\hat{P}^I_{MK}$ annihilates all components $|\Phi_{K'}\rangle$ with $K\neq K'$, we get

$$|\Psi^{IM}\rangle = \sum_K \hat{P}^I_{MK} \sum_{K'} g_{K'}|\Phi_{K'}\rangle = \sum_K \hat{P}^I_{MK}|\tilde{\Phi}\rangle. \tag{11.130}$$

The function

$$|\tilde{\Phi}\rangle = \sum_K g_K|\Phi_K\rangle \tag{11.131}$$

will, in general, no longer be a product wave function. However, there are

many cases in which it can be approximated by a function of this type.* In all such cases, the variation of g_K is redundant if we carry out a general variation of $|\Phi\rangle$. We then end up with the simple operator (originally proposed by Kamlah [Ka 68]):

$$\hat{P}^I_M = \sum_K \hat{P}^I_{MK}. \tag{11.132}$$

It has the property $\hat{P}^2 = \hat{P}$, but is not Hermitian. It also violates rotational symmetry [So 77] in the sense that it is not invariant under a rotation of the function $|\tilde{\Phi}\rangle$. Together with a full variation of $|\tilde{\Phi}\rangle$, however, the procedure stays rotationally invariant, because we always find $|\tilde{\Phi}\rangle$ at such an orientation that the energy has a minimum. By symmetry arguments, the quantization axis determined by the operator $\hat{P}^I_M$ should coincide with one of the principal axes of the mass distribution in $|\tilde{\Phi}\rangle$.

The operator (11.132) annihilates all components in the wave function $|\tilde{\Phi}\rangle$ that do not have the signature

$$r_x = (-)^I. \tag{11.133}$$

In even nuclei, this fact guarantees a spin sequence $I = 0, 2, 4, \ldots$ in the ground state band, even if it is no longer axially symmetric. In odd nuclei it causes the $\Delta I = 2$ level sequence in favored and unfavored bands to have a different signature (see Sec. 3.3).

11.4.6.3. The Self-Consistent Cranking Model. In Section 11.4.4, we presented a method of deriving an approximate expression for the projected energy without specifying much about the internal wave function $|\Phi\rangle$. For pedagogical reasons, it was discussed in that section for the case of a one-dimensional rotor. It can also be applied to the general case of three-dimensional rotations [Ka 68, BMR 70, Ma 75a].

Since the three-dimensional case involves rather lengthy calculations, we give here only the main result: Under the following three assumptions for the "internal" wave function $|\Phi\rangle$ namely:

(i) strong deformation ($\langle\Phi|\hat{J}_y^2|\Phi\rangle \gg 1$),

(ii) definite signature ($e^{i\pi\hat{J}_x}|\Phi\rangle \propto |\Phi\rangle$),

(iii) approximate axial symmetry for small I-values (i.e., in the ground state band $\langle\Phi|\hat{J}_z^2|\Phi\rangle \ll I(I+1)$),

we find up to second order in the Kamlah expansion (11.80) the following

*A simple example is a wave function of an odd nucleus

$$|\Phi\rangle = \sum_K C_K \beta_K^+ |\Phi_0\rangle,$$

where an arbitrarily strongly mixed particle sits on a core with $K=0$; another example is a wave function $|\Phi\rangle$ which deviates only slightly from axial symmetry.

expression for the projected energy* [BMR 70]

$$E^I_{\text{proj}} = \frac{\langle H\hat{P}^I_M\rangle}{\langle \hat{P}^I_M\rangle} = \langle H\rangle - \frac{\langle \Delta\hat{\mathbf{J}}^2\rangle}{2\vartheta_Y} + \omega\left(\sqrt{I(I+1)-\langle \hat{J}_z^2\rangle} - \langle \hat{J}_x\rangle\right)$$
$$+ \frac{1}{2\vartheta_Y}\left(\sqrt{I(I+1)-\langle \hat{J}_z^2\rangle} - \langle \hat{J}_x\rangle\right)^2, \tag{11.134}$$

with

$$\frac{1}{2\vartheta_Y} = \frac{\langle (H-\langle H\rangle)\Delta\hat{\mathbf{J}}^2\rangle}{2\left(\langle \Delta\hat{J}_x^2\rangle^2 + \langle \hat{J}_y^2\rangle^2 + \langle \hat{J}_z^2\rangle^2\right)}, \qquad \omega = \frac{\langle H\Delta\hat{J}_x\rangle}{\langle \Delta\hat{J}_x^2\rangle}.$$

These expressions are very similar to Eq. (11.91), and we can draw the same conclusions: In the case where the term $\langle\Delta\hat{\mathbf{J}}^2\rangle/2\vartheta_Y$, which removes the spurious contributions in the energy, can be neglected in the variation, we end up with the cranking model

$$\delta\langle H-\omega\hat{J}_x\rangle = 0 \tag{11.135}$$

with the subsidiary condition

$$\langle \hat{J}_x\rangle = \sqrt{I(I+1)-\langle \hat{J}_z^2\rangle}\ . \tag{11.136}$$

It differs from the usual constraint (7.97) by the term $\langle \hat{J}_z^2\rangle$. In the ground state band of well-deformed nuclei, it does not play any role: For small I-values the nucleus is nearly axially symmetric with $K=0$, and for large angular momenta it can be neglected anyway. However, it is important to take it into account for cranking calculations in $K\neq 0$ bands, particularly in odd A nuclei [RMB 74, RM 74].

We thus have a microscopic derivation of the cranking model (Sec. 3.4) which was originally introduced by classical arguments. The cranking wave functions have to be understood as internal wave functions. The wave function in the laboratory frame is obtained by an angular momentum projection with the operator (11.132). This derivation is only based on the assumption of large particle numbers and large deformations, which guarantee a sharply peaked overlap function $n(\Omega)$. In that sense, it is a fully quantum mechanical derivation, which does not depend on classical arguments such as large angular momenta with $\langle\Delta\hat{\mathbf{J}}^2\rangle \ll I^2$; on the contrary, the derivation is based on the fact that we have large fluctuations $\langle\Delta\hat{\mathbf{J}}^2\rangle \gg 1$ (narrow overlap). This means also via the uncertainty relation that the spread in angle is small ($\langle\Delta\Omega^2\rangle \ll 1$).

In Chapter 12 we will rederive these relations in a time-dependent formulation with wave packets. For example, in the case of translations it will also turn out that the physically realized fact that nuclei are rather

* $\hat{P}^I_M$ is the operator defined in Eq. (11.132), and the wave function $|\Phi\rangle$ is replaced by a simple ket $|\ \rangle$.

localized in space ($\langle \Delta \mathbf{R}^2 \rangle^{1/2}$ very small compared to the nuclear radius, that is, a small width of the wave paket) has the consequence that the momentum spread $\langle \Delta \mathbf{P}^2 \rangle$ is large. Therefore, the self-consistent cranking model (pushing model in this case) will turn out to be a good approximation.

11.4.6.4. Electromagnetic Moments and Transition Probabilities. In Section 11.4.4 we saw that for the calculation of rotational spectra in well-deformed nuclei, we do not need an explicit angular momentum projection. The cranking model already provides a rather good approximation. The same is no longer true for matrix elements of operators, which do not behave like a scalar under rotations, such as electromagnetic multipole operators $\hat{Q}_{\lambda\mu}$.

For the reduced matrix element of the operator $\hat{Q}_{\lambda\mu}$ between normalized wave functions $|\Psi_1^{I_1 M_1}\rangle$ and $|\Psi_2^{I_2 M_2}\rangle$ of the type (11.117), we obtain

$$\langle \Psi_1^{I_1} \| \hat{Q}_\lambda \| \Psi_2^{I_2} \rangle = \sum_{\substack{K_1 K_2 \\ \bar{K}_2 \mu}} g_{K_1}^{(1)*} g_{K_2}^{(2)} (-)^{I_1 - K_1} \begin{pmatrix} I_1 & \lambda & I_2 \\ -K_1 & \mu & \bar{K}_2 \end{pmatrix} ((2I_1+1)(2I_2+1))^{1/2} \cdot Q_{\lambda\mu}^{\text{intr}}(K_2, \bar{K}_2), \tag{11.137}$$

with

$$Q_{\lambda\mu}^{\text{intr}}(K_2 \bar{K}_2) = \mathfrak{N}_1 \mathfrak{N}_2 ((2I_1+1)(2I_2+1))^{1/2} \times \frac{1}{8\pi^2} \int d\Omega \, D_{\bar{K}_2 K_2}^{I_2}(\Omega) \langle \Phi_1 | \hat{Q}_{\lambda\mu} R(\Omega) | \Phi_2 \rangle. \tag{11.138}$$

The factors $\mathfrak{N}_i$ are normalization constants (i refers to the intrinsic wave function $|\Phi_i\rangle$)

$$\mathfrak{N}_i^{-2} = \sum_{KK'} g_K^{*(i)} g_{K'}^{(i)} n_{KK'}^{I_i (i)}. \tag{11.139}$$

To evaluate the overlap integrals $\langle \Phi_1 | \hat{Q}_{\lambda\mu} R(\Omega) | \Phi_2 \rangle$, we can again use a sort of Kamlah expansion (11.80) and obtain, in zeroth order,

$$\langle \Phi_1 | \hat{Q}_{\lambda\mu} R(\Omega) | \Phi_2 \rangle = \langle \Phi_1 | \hat{Q}_{\lambda\mu} | \Phi_2 \rangle \langle \Phi_2 | R(\Omega) | \Phi_2 \rangle. \tag{11.140}$$

Using this approximation, we find

$$Q_{\lambda\mu}^{\text{intr}}(K_2 \bar{K}_2) \propto \langle \Phi_1 | \hat{Q}_{\lambda\mu} | \Phi_2 \rangle \int d\Omega \, D_{\bar{K}_2 K_2}^{I_2}(\Omega) n_2(\Omega). \tag{11.141}$$

The remaining integrals have to be carried out explicitly. Only in the limit of very large deformations do we obtain [Vi 66, Ze 67]

$$Q_{\lambda\mu}^{\text{intr}}(K_2 \bar{K}_2) = \langle \Phi_1 | \hat{Q}_{\lambda\mu} | \Phi_2 \rangle \delta_{K_2 \bar{K}_2} \tag{11.142}$$

and

$$\langle \Psi_1^{I_1} \| \hat{Q}_\lambda \| \Psi_2^{I_2} \rangle = \sum_{K_1 K_2 \mu} g_{K_1}^{(1)*} g_{K_2}^{(2)} ((2I_1+1)(2I_2+1))^{1/2} (-)^{I_1 - K_1} \begin{pmatrix} I_1 & \lambda & I_2 \\ -K_1 & \mu & K_2 \end{pmatrix} \times \langle \Phi_1 | \hat{Q}_{\lambda\mu} | \Phi_2 \rangle, \tag{11.143}$$

a result which corresponds to Eq. (1.70) in the collective model. The only difference is that the intrinsic multipole moments can now be calculated microscopically. In

particular, we take into account differences between the intrinsic wave functions $|\Phi_1\rangle$ and $|\Phi_2\rangle$ by the overlap integral $\langle\Phi_1|\hat{Q}_{\lambda\mu}|\Phi_2\rangle$.

In cases where this first approximation is not good enough, we need explicit expressions for the norm overlap $n(\Omega)$ in the three-dimensional case. It can be expanded as [IMR 79]

$$n(\Omega)=\exp\Big(\sum_{LKK'} C^L_{KK'} D^{L*}_{KK'}(\Omega)\Big). \tag{11.144}$$

The Gaussian approximation corresponds to a restriction to $L\leqslant 2$. In analogy to Eq. (11.79), we can express the coefficients $C^L_{KK'}$ by expectation values of $\langle\hat{J}_i\rangle$ and $\langle\hat{J}_i\hat{J}_k\rangle$. Using the relations (A.17), for large deformations and small K-admixtures [BMR 70, VS 71] we get

$$n(\Omega)\simeq\exp\left\{-\frac{\langle\hat{J}_y^2\rangle}{2}\beta^2+(\cos(\alpha+\gamma)-1)\langle\hat{J}_z^2\rangle+\frac{i}{2}\langle\hat{J}_x\rangle\beta(\sin\alpha-\sin\gamma)\right\}. \tag{11.145}$$

11.4.6.5. Exact Angular Momentum Projection. Calculations with exact angular momentum projected wave functions usually require a large numerical effort. The expectation value of each operator—for instance, the Hamiltonian—contains a sum over a series of three-dimensional integrals [see Eqs. (11.118) and (11.119)]. The integrands are proportional to the overlap functions $\langle\Phi|R(\Omega)|\Phi\rangle$ and $\langle\Phi|HR(\Omega)|\Phi\rangle$, which have to be calculated point by point using the formula of Onishi (E.49). Only in cases of axially symmetric intrinsic wave functions (with the quantum number K) do the α- and γ-dependences become trivial, and we end up with a one-dimensional integral:

$$\langle\Psi^{IM}|H|\Psi^{IM}\rangle=\frac{\int d\cos\beta d^I_{KK}(\beta)\langle\Phi|e^{i\beta\hat{J}_y}H|\Phi\rangle}{\int d\cos\beta d^I_{KK}(\beta)\langle\Phi|e^{i\beta\hat{J}_y}|\Phi\rangle}.$$

In this form, the angular momentum projection before or after the variation has been applied very often in the literature (for a review, see [Ri 68, Ma 70, Wo 75]). Many investigations have been carried out within the $1p$ and the $2s-1d$ *shell*, where we have deformed nuclei with spectra of rotational character [BBC 72]. In this region of the periodic table the angular momentum projected HF theory can be compared with exact shell model calculations [Re 58, KP 59, Wa 70b, MMW 73], and in many cases we find a perfect agreement. In detail, the projected HF theory has been applied in many versions in these light nuclei. In the simplest case one often projects after variation (see Sec. 11.4.2) from axially symmetric deformed product states [KL 64, BGR 65, WG 67b, GFW 72]. An improvement is obtained by a projection before variation with intrinsic wave functions that depend on several variables [BLD 67, DFG 68, ABC 69, LB 68, 69]. Sometimes one must also take pairing correlations into account. This is usually done in the framework of particle number projected BCS or HFB theory [GGF 73, GGA 74]. To treat further correlations, some authors finally have also superimposed several angular momentum projected wave functions [TG 69, CP 70, Da 70, Wa 71, MKL 71] in the sense of the GCM ansatz of Chapter 10.

In the $2s-1d$ shell of some nuclei, triaxial HF solutions have been found. In such cases one must carry out a full three-dimensional projection. The ansatz (11.112), however, seems to be too complicated for actual applications. So far, in all

practical calculations other representations of the projection operator have been used [Un 63, Sh 65, Ke 66, GS 69, Ra 70, GM 71, CL 73, MWW 74, Wa 74, KD 75, 76] which, however, seem to be restricted to applications in light nuclei.

In *heavy nuclei*, an exact angular momentum projection has so far only been carried out in axially symmetric cases. There, however, the method again yields very promising results for the ground state properties and for the low-lying members of the ground state rotational band* in deformed nuclei [OSY 70, AN 71].

We can conclude this section with the remark that projection methods in connection with the self-consistent field approach seem to be a very powerful tool for the description of collective rotational motion, though in detail they can be rather complicated.

11.4.7 The Structure of the Intrinsic Wave Functions

Until now, we have only investigated the problem of how to construct symmetry-conserving wave functions $|\Psi^I\rangle$ from the intrinsic wave function $|\Phi\rangle$. We could also ask the opposite question: Is it possible to represent the symmetry-violating wave function $|\Phi\rangle$ as a superposition of the functions $|\Psi^I\rangle$, and how large are the components with different I values, that is to calculate the spread $\langle \Delta \mathbf{J}^2 \rangle$?†

In the case of Abelian groups, the operator P^I (11.57) is a real projector. If $|I\alpha\rangle$ is a complete orthogonal set, we find

$$|\Phi\rangle = \sum_{I\alpha} C_{I\alpha} |I\alpha\rangle, \tag{11.146}$$

and the probability of finding a component with eigenvalue I in the symmetry violating function $|\Phi\rangle$ is given by

$$W_I = \sum_{\alpha} |C_{I\alpha}|^2 = \langle \Phi | \hat{P}^I | \Phi \rangle = \frac{1}{2\pi} \int d\alpha \, e^{-iI\alpha} n(\alpha), \tag{11.147}$$

where W_I is the Ith Fourier component of the norm overlap $n(\alpha)$. If $|\Psi^I\rangle$ is normalized, we can write

$$|\Phi\rangle = \sum_{I} \sqrt{W_I} \, |\Psi^I\rangle. \tag{11.148}$$

In the case of three-dimensional rotations, we can again decompose

$$|\Phi\rangle = \sum_{IM\alpha} C_{IM\alpha} |IM\alpha\rangle. \tag{11.149}$$

The probability of finding an eigenstate with eigenvalue $I(I+1)$ in the function Φ is given by [GSF 78a, IMR 79]:

$$W_I = \sum_{\alpha M} |C_{IM\alpha}|^2 = \sum_{M} \langle \Phi | \hat{P}^I_{MM} | \Phi \rangle = \sum_{M} n^I_{MM}. \tag{11.150}$$

* For the description of excited states of vibrational character, a TDA approach based on projected two-quasi-particle states seems to be a very powerful method [OS 69, KI 70, AG 74, HIT 79].

† An early investigation of this type has been carried out in [AY 59].

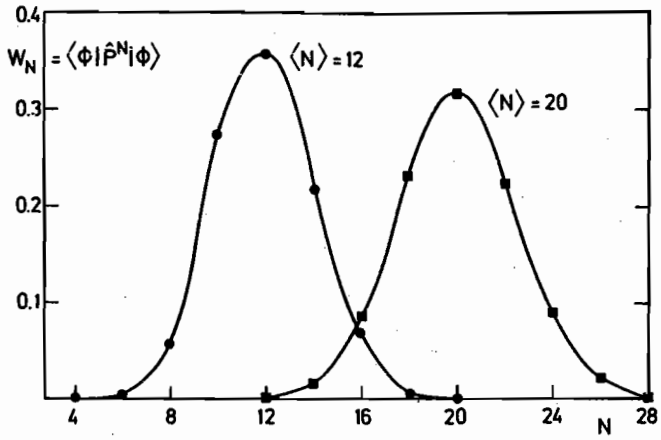


Figure 11.6. Probability components with N particles in the BCS ground state wave function with average numbers $\langle N\rangle = 12$ and $\langle N\rangle = 20$ for realistic nuclei in the rare earth region.

Figure 11.6 shows the decomposition of two BCS wave functions into their components having eigenvalue N for the *particle number*. They differ in the average value $\langle N\rangle$. We see that the violation of the particle number is rather weak. The distribution corresponds roughly to a Gaussian with a relatively small $\Delta N^2 \simeq 4$–6. It is therefore crucial to conserve the particle number at least on the average ("cranking" approach). Consequently, the first approximation in the Kamlah expansion (11.80) is often not sufficient. In many cases an exact projection is important.

The situation is quite different in the case of *angular momenta*. Figure 11.7 shows the components W_I [Eq. (11.150)] of different angular momenta in the internal wave function of the well-deformed nucleus ^{164}Er, calculated by the self-consistent cranking model for different values of $\langle\hat{J}_x\rangle$. We

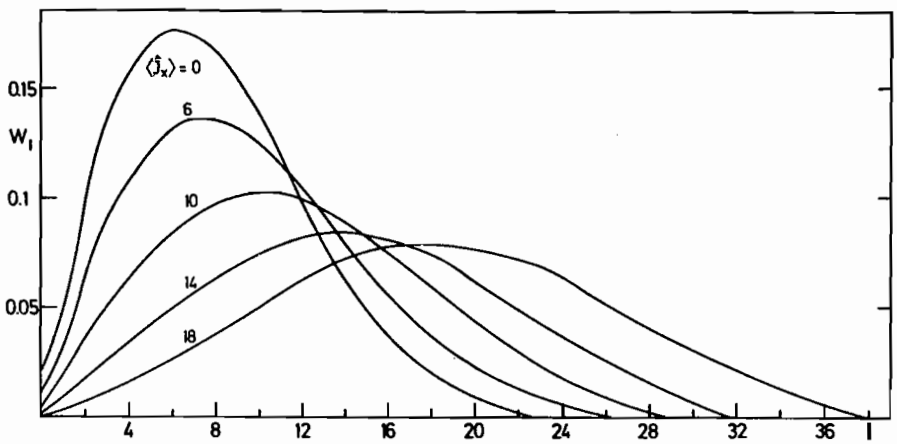


Figure 11.7. Probability components with angular momentum I in the HFB wave function of ^{164}Er for different values of $\langle\hat{J}_x\rangle = \sqrt{I(I+1)}$. (From [IMR 79].)

now have a strong symmetry violation: $\langle \Delta \hat{\mathbf{J}}^2 \rangle \simeq 100$, this corresponds to a spread in angle of roughly 10°, a rather sharp distribution indeed. In the ground state wave function $I=0$ is represented only by a relatively small component. The maximum lies at $I \simeq 6$. The reason for this unexpected behavior is the fact that angular momenta are not additive quantities, and there is a much higher number of possibilities, in the configuration space under consideration, of coupling to $I=6$ than to $I=0$. Since we are interested in taking into account as many correlations as possible in the form of a generalized product state, we have to allow for such large spurious components. The projection puts all these components to zero. Only in the limit of high spins does the maximum of the distribution lie at the average value of $\hat{J}_x$.